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PrivaFed: A Privacy-Preserving Federated Learning Framework

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Abstract

Federated Learning (FL) has emerged as a promising paradigm for distributed model training without requiring centralized access to sensitive data. In contrast to conventional methods of Machine Learning (ML), FL can be used with hundreds of clients, including hospitals, research centers, and edge devices, to train models locally. Crucially, only the model updates are shared and combined to improve global performance. This preserves privacy of data, but its real-world applications face challenges such as incomplete and imbalanced data, communication overhead, and scalability. These challenges highlight the need for personalized and lightweight solutions. This project aims to design a practical FL framework combined with Transfer Learning (TL) and lightweight techniques that addresses these challenges while ensuring efficiency and privacy. This report introduces "PrivaFed: A Privacy-Preserving Federated Learning Framework". This framework is specifically tailored for Brain Magnetic Resonance Imaging (MRI) in stroke lesion analysis, a domain selected because of its societal importance and the existence of publicly available datasets.

Executive Summary

PrivaFed: A Privacy-Preserving Federated Learning Framework is a project that deals with one of the most significant issues in the field of healthcare Artificial Intelligence (AI), which is how to collaboratively train powerful and reliable Deep Learning (DL) models on sensitive patient data while adhering to strict privacy standards. The framework is specifically designed for the task of stroke lesion segmentation in brain MRI scans, which is very crucial in making clinical diagnoses, treatment plans, and long-term monitoring of the patients.

Neurological disorders were the most common cause of illness in 2021, impacting 3.4 billion individuals, or more than 4 out of 10 persons in the world. Stroke is among the neurological disorders that are major causes of death around the world. Timely and precise segmentation of lesions in MRI scans is therefore crucial for implementing successful medical interventions. Although DL has demonstrated impressive performance in medical image analysis, such models require large and heterogeneous datasets. In practice, privacy laws and morality significantly limit the data sharing between institutions, and therefore, centralized training solutions are not feasible.

FL offers a promising solution to this by enabling a number of institutions to jointly train models without sending patient data to a central server. Instead, it is done locally within each institution, and model parameters or updates are only shared for secure aggregation. This guarantees confidentiality of data but introduces new challenges of non-independent and imbalanced datasets, communication overhead, and resource constraints across institutions involved in the process.

PrivaFed is designed to overcome these obstacles by using a privacy-sensitive FL model. The system uses TL to focus on the use of pre-trained models to improve performance when the recent datasets are limited. The system also utilizes lightweight approach and these methods reduce the computing resources and increase the speed of convergence. Moreover, the framework focuses on scalability and practicality, making it adaptable to real-world healthcare environments with a variety of infrastructures.

The project makes three contributions by addressing privacy, efficiency, and clinical applicability. It ensures privacy preservation by keeping sensitive medical records private and preventing raw data from leaving the institution. It is robust, reducing communication and computational expenses while maintaining a high degree of segmentation performance on MRI scans. Lastly, it holds strong clinical relevance, fulfilling a critical medical need by focusing on stroke lesion detection, with potential expansion to other neurological conditions.

Ultimately, PrivaFed is one step towards enhancing secure, decentralized, and collaborative AI in healthcare, in which institutions can gain the benefits of shared knowledge without losing patient confidential-

ity. The presented project shows that FL with TL and a personalized approach can fill the gap between data privacy and advanced medical image analysis and offer a scalable and effective solution in the future of digital healthcare.

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Chapter 1 Introduction

The title “PrivaFed: A Privacy-Preserving Federated Learning Framework” highlights two key aspects of our project: privacy and FL. The name reflects the core idea of developing a secure and distributed framework that enables collaborative training of segmentation models across healthcare institutions without sharing sensitive patient data. By focusing on stroke lesion segmentation in MRI scans, PrivaFed addresses a critical clinical need while ensuring confidentiality and efficiency in real-world medical environments.

Stroke is one of the major causes of disability worldwide, and segmentation of stroke lesions in MRI scans is vital for diagnosis and treatment planning. Though the centralized approaches have shown good results in the sphere of medical imaging but they are constrained by privacy issues and the scarcity of data. FL provides an escape as it allows model training to be performed across institutions without them sharing raw data. Nevertheless, the issue of heterogeneous data sets, limited resources, and communication overhead persists.

The proposed project will deploy a privacy preserving framework which combines FL with TL to enhance segmentation performance over diverse and distributed data sets. PrivaFed aims to support healthcare institutions through the provision of real-time, secure, lightweight, and scalable stroke lesion segmentation solutions.

1.1 Purpose of this Document

The purpose of this document is to present the design, development, and evaluation of PrivaFed, a privacy-preserving FL framework for stroke lesion segmentation in brain MRI scans. The main goal of the research is to determine whether FL can be effectively applied to build an efficient and personalized segmentation model while ensuring patient data privacy.

The increasing demand for AI-based medical imaging solutions that respect patient privacy while efficiently handle distributed healthcare data stems the motivation behind this project. Centralized training methods raise serious concerns about data confidentiality and potential violations of institutional policies. To address these challenges, this work focuses on creating a privacy-preserving framework for stroke lesion segmentation.

The study explores the practical application of FL for distributed medical data analysis and integrates advance model techniques such as TL to enhance segmentation performance across diverse datasets. It aims to help healthcare institutions utilize real-time patient data securely and effectively.

The central research question guiding this project is: Can FL, combined with TL, enable a lightweight and privacy-preserving framework for stroke lesion segmentation in brain MRI scans? This report provides the background of the problem, outlining the proposed solution, explaining the methodology, presenting results, and discussing the significance and limitations of the findings.

1.2 Intended Audience

The intended audience for this Final Year Project (FYP) include:

- Healthcare practitioners who require assistance in lesion segmentation as a second opinion for clinical decision-making and treatment planning.
- Healthcare institutions, including hospitals, clinics, and research centers that work with distributed medical data under strict privacy constraints.
- Researchers and scientists in the medical imaging, neurology and ML community that are studying federated methods to segmentation tasks.
- The academic review board comprising of professors and examiners who will examine this FYP based on its technicality, novelty, and applicability to real world problems.

By addressing the needs of these diverse stakeholders, this report ensures that the technical contributions and practical implications of the project are conveyed to the readers.

1.3 Definitions, Acronyms, and Abbreviations

All important definitions, acronyms, and abbreviations used in this document are listed below.

1.3.1 Definitions

Brain Stroke: Medical condition that arises as a result of reduced blood supply to a part of the brain.

Stroke Lesion: The damaged brain tissue visible in MRI scans after a stroke.

Lesion Segmentation: The process of automatically identifying and outlining stroke-affected regions in MRI scans.

Federated Learning: A distributed learning approach where multiple clients collaboratively train a model without having to share raw data.

Transfer Learning: A machine learning technique in which knowledge learned from a task is re-used in order to improve performance on a related task.

1.3.2 Acronyms and Abbreviation

FL: Federated Learning

ML: Machine Learning

AI: Artificial Intelligence

FYP: Final Year Project

TL: Transfer Learning

DL: Deep Learning

MRI: Magnetic Resonance Imaging

1.4 Conclusion

To conclude, this chapter highlights the goals, motivation, and significance of PrivaFed with reference to medical imaging and data privacy. To address challenges such as privacy risks, imbalanced datasets, scarcity of data, and communication overhead, PrivaFed focuses on the integration of FL and TL techniques. The following chapters elaborate on the concepts introduced here, covering literature review, methodology, implementation, evaluation, and conclusions.

Chapter 2 Project Vision

This chapter discusses the project vision, with emphasis on the need for a privacy-preserving stroke segmentation framework. The goals, scope, and sustainable development milestones PrivaFed aims to achieve are listed below.

2.1 Problem Domain Overview

“PrivaFed: A Privacy-Preserving Federated Learning Framework” focuses on advanced brain stroke segmentation through a federated setup, using ML to analyze brain scan data. The system trains models in a distributed system with brain imaging data to determine the segmentation of stroke lesions. To avoid aggregation of sensitive medical data at a central server, the federated approach is used, through which only model updates are shared with the central server while the patient data remains at the local server. This makes the system privacy-preserving and suitable for real healthcare scenarios. Data imbalance, differences in client resources, and communication costs are some of challenges handled by the framework. PrivaFed’s aim is to provide a secure, efficient, and lightweight second opinion to medical practitioners in stroke segmentation through the integration of TL and personalized training.

2.2 Problem Statement

The practical effectiveness of combining FL techniques with TL and lightweight approach for stroke analysis on brain scans remains an open challenge. Current methods use centralized data, which is not feasible and raises serious privacy issues in medical domain. This gap in brain health segmentation inspired the work for PrivaFed. To handle this issue, PrivaFed is designed to preserve the privacy of data and minimize the computation cost to make a more practical collaborative healthcare AI for practical deployment.

2.3 Problem Elaboration

The approaches used by current stroke segmentation models are practically inadequate for deployment in real-life medical environment. Their major limitations are outlined below:

- **Privacy Risks:** The current standard centralized training approaches need to pool patient scans in one location. This is a threat to the privacy of sensitive medical data.
- **Lack of Transfer Learning:** Majority of the systems train segmentation models from scratch which is wasteful given a very small size of dataset. Therefore, using TL, performance can be

optimized and training time can be shortened. Nevertheless, TL is rarely investigated in federated framework for stroke segmentation, highlighting a potential research gap.

- **Demand for Lightweight Models:** Most DL frameworks are computationally expensive, resulting in longer training times and increased communication overhead. However, real healthcare settings have limited hardware, hence building lightweight solutions is necessary.
- **Handling Data Imbalance:** Stroke lesion data is often heterogeneous, being scarce and inhomogeneously distributed across sites, therefore in the absence of effective means that can account for this imbalance the models obtained may not adequately perform on critical cases.
- **Lack of Personalization:** A single global model may not work well with variations in patient populations or institutional data. The model's utility in clinical applications is therefore limited.

These issues demonstrate the need to introduce a new framework that maintains privacy by being decentralized, integrates TL, applies lightweight models, and supports personalization for more accurate and practical stroke segmentation.

2.4 Goals and Objectives

- **Privacy Preservation:** Create a secure FL framework to allow collaborative model training across many clients without sharing the raw data.
- **Personalization:** Address the issue of heterogeneous and imbalanced datasets through a solution that incorporates adaptive, personalized FL functionality.
- **Lightweight:** Build lightweight model to minimize the communication and computational costs in actual healthcare systems.
- **Optimized Model Updates:** Improve training efficiency through optimized model updates.
- **Transfer Learning:** Use TL strategies to use pretrained models to minimize training time and improve model accuracy especially when dealing with small medical imaging datasets.

2.5 Project Scope

The project is divided into two main phases: the research phase and the development phase.

2.5.1 Research Phase

- Carry out a comprehensive literature review to gather insights, identify challenges, and study best practices that have been taken by prior researchers on stroke segmentation and privacy-sensitive AI.
- Study various techniques in TL, model compression, and class imbalance handling to assess their suitability for this research.
- Identify and evaluate open-source brain imaging datasets that can be used for model training and testing.
- Examine effective preprocessing techniques to make the dataset compatible with the model, and define metrics to assess model performance.

2.5.2 Development Phase

- Collect and prepare diverse brain scan datasets in order to create ML models that are able to perform reliable stroke segmentation in FL framework.
- Apply TL to use pretrained models for faster convergence and better performance on data.
- Design and implement lightweight architectures that reduce computational and communication costs.
- Apply different techniques to handle data imbalance and heterogeneity.
- Adopt optimized model update strategies in order to improve the performance when training the federated model.
- Conduct experiments including ablation experiments as a test to assess the effect of privacy, TL and personalization on overall performance.
- Design an easy to use interface that allows healthcare professionals to upload brain scan data and view segmentation results in real time.

2.6 Sustainable Development Goal (SDG)

The development goal of our project aligns with SDG 3: Good Health and Well-Being, as shown in Figure 2.1. It aims to demonstrate how FL can serve as a practical and scalable solution for stroke segmentation. By countering key challenges such as data privacy, limited computational power, and differences in medical datasets, PrivaFed enhances healthcare outcomes and promotes overall well-being.



Figure 2.1: Sustainable Development Goal 3

2.7 Constraints

During model development and testing, limited access to diverse and high-quality brain scan datasets may pose as a significant constraint. The availability of sufficient computational resources is also a challenge, as training and evaluating DL models require high-performance hardware. Finally, the scale of experimentation and validation can be restricted by the time limitation of this FYP.

2.8 Conclusion

In conclusion, PrivaFed envisions a privacy friendly and effective structure of brain stroke segmentation through FL. PrivaFed achieves this vision by integrating TL, lightweight architectures and customized modeling to overcome the obstacles of data privacy, heterogeneity, and resource limitations. PrivaFed is a contributor towards real-world and collaborative AI in the field of healthcare according to SDG 3.

Chapter 3 Literature Review

Before developing the proposed solution, it is essential to explore existing research in the field and related areas to build a clear understanding of the problem and its potential solutions. This process also helps establish a strong knowledge base to guide the design of a privacy-preserving federated framework. The literature review is organized into three parts: brain stroke segmentation, TL and model generalization strategies in brain stroke segmentation, and FL in brain stroke segmentation.

3.1 Definitions, Acronyms, and Abbreviations

3.1.1 Definitions

F1-score: Harmonic mean of Precision and Recall

Tmax: A parameter that measures how delayed the blood flow is in a specific region of the brain compared to normal tissue.

T1-weighted MRI: A specific type of MRI scan that highlights certain tissue characteristics based on their T1 relaxation time.

Acute: Early stage of stroke.

Subacute: Intermediate stage of stroke.

Chronic: Late stage of stroke.

3.1.2 Acronyms and Abbreviations

U-Net: U-shaped Network

nnU-Net: no new U-Net

ResNet: Residual Network

CNN: Convolution Neural Network

DWI: Diffusion-weighted MRI

FLAIR: Fluid-Attenuated Inversion Recovery

CT: Computed Tomography

ADC: Apparent Diffusion Coefficient

DSC: Dice similarity coefficient

AHD: Average Hausdorff Distance

N4: Nonparametric Nonuniform intensity Normalization version 4

MONAI: Medical Open Network for AI

HCSNet: Hybrid Contextual Semantic Network

SQMLP-Net: Simulated Quantum Mechanics-based Multi-Layer Perceptron Network

HD95/HD: Hausdorff Distance

ASSD: Average Surface Distance

GDFL: Generalized Dice Focal Loss

IoU: Intersection Over Union

SMOTE: Synthetic Minority Oversampling Technique

Non-IID: Non-Independent and Identically Distributed

3.2 Detailed Literature Review on Brain Stroke Segmentation

This section provides detailed information about the research that has been done by different researchers on Brain Stroke Lesion Segmentation, which will be helpful for our research.

3.2.1 CNN-Res: deep learning framework for segmentation of acute ischemic stroke lesions on multimodal MRI images

3.2.1.1 Summary of the research item

This study by Gheibi et al. introduced a deep convolution neural network “CNN-Res”, designed to automatically segment acute ischemic stroke lesions from multimodal MRI images, Diffusion-Weighted Imaging (DWI), and Fluid-Attenuated Inversion Recovery (FLAIR) in particular [1]. The model combines the strengths of U-shaped Network (U-Net) and Residual Neural Network (ResNet) architectures in a U-shaped encoder–decoder framework. Its performance was enhanced with residual and translation blocks to reduce gradient degradation and computational cost. The bottleneck layer in the architecture compresses feature maps, enabling faster training with a small parameter count, approximately 6.4 million, which is much less than the U-Net’s 31 million. Data was obtained from the Neuroscience Center of Tabriz University of Medical Sciences, where MRI scans from 44 stroke patients were preprocessed, annotated by neurologists, and augmented to overcome limited training samples. CNN-Res achieved a Dice similarity coefficient (DSC) of 85.43% on the local dataset and 79.23% on the public SPES 2015 dataset, and thereby outperformed the standard U-Net by achieving competitive accuracy with less computational complexity.

3.2.1.2 Critical Analysis

Strengths

- Combines U-Net’s spatial precision with ResNet’s residual learning to suppress vanishing gradient problems.

- A smaller parameter count of approximately 6.4 million and shorter training time compared to the U-Net result in low computational cost.
- Achieved high DSC values of 85.43% on the local dataset and 79.23% on the public dataset.
- Use of multi-modal MRI (i.e., DWI and FLAIR) improved lesion delineation and provided robustness to image variations.
- Comprehensive preprocessing and expert annotation ensured data quality and accurate ground-truth masks.

Weaknesses

- Model processes 2D slices rather than full 3D MRI volumes, which can cause it to miss the spatial continuity of lesions.
- Model's performance may vary across scanners or patient populations, as no federated or multi-site testing was performed.
- Centralized training on local data may not suit large-scale clinical environments.
- Model relies on expert labeling, which may introduce subjectivity.

3.2.1.3 Relationship to the proposed research work

The CNN-Res framework provides a strong architectural foundation for ischemic stroke lesion segmentation but operates in a centralized and data limited setting. In contrast, PrivaFed framework aims to extend this concept by enabling federated and privacy preserving training across multiple healthcare institutions. While CNN-Res highlights the importance of multimodal input and efficient residual learning for accurate segmentation, through FL and TL, PrivaFed will address its major limitations, which include data scarcity, domain generalization, and privacy concerns. By distributing model training across diverse, non-IID datasets without sharing patient data, PrivaFed aims to achieve comparable or even better segmentation accuracy while maintaining compliance with healthcare data protection standards.

3.2.2 Generalizable automated ischaemic stroke lesion segmentation with vision transformers

3.2.2.1 Summary of the research item

The paper presents a transformer based DL framework designed for the automated segmentation of ischemic stroke lesions in DWI scans [2]. The paper handles the challenge of developing a model that not only achieves high segmentation accuracy but also generalizes across diverse clinical settings. In

order to achieve this, the authors have utilized "Swin UNETR" architecture which being a vision transformer model can capture both fine local details and global contextual information. The framework was trained and assessed using a substantial, internal dataset gathered from multiple sites, consisting of 3,563 manually identified stroke lesions and 6,900 normal control cases with extensive data curation and augmentation. The model demonstrated an average Dice Similarity Coefficient (DSC) of approximately 0.89 and a Hausdorff Distance (HD) of 2.6 voxels. Comparisons show that the model performed better than the conventional U-Net baseline. Although external datasets were used to check lesion distribution, formal evaluation was limited to the internal data because expert annotations were not available in external datasets.

3.2.2.2 Critical Analysis

Strengths

- The model maintains high segmentation accuracy on multiple external datasets without retraining, demonstrating strong generalization.
- Use of transformer architecture captures long-range spatial dependencies, allowing better contextual understanding than traditional Convolution Neural Network (CNN).
- Model achieves high segmentation accuracy with DSC around 0.89 and reduced boundary errors.
- Model employs a threshold average loss and an iterative curation strategy that improves model precision and reduces false positives.

Weaknesses

- Computationally heavy models as vision transformers require high-end GPUs and large memory, limiting use in resource-constrained hospitals.
- The model under performs on tiny infarcts, similar to most DL models.
- Due to centralized training, all data was combined in the process, compromising privacy and institutional data separation.
- External validation was restricted to unlabeled datasets; hence, true out-of-distribution quantitative performance could not be conclusively established.

3.2.2.3 Relationship to the proposed research work

This study provides a valuable foundation for PrivaFed, as it highlights the effectiveness of transformer-based architectures for stroke lesion segmentation while simultaneously exposing the limitations of cen-

tralized learning. The research results point out that high-performing models can generalize well across heterogeneous MRI data. Simultaneously, it also demonstrates the essential need for privacy-preserving approaches when handling sensitive medical datasets from multiple sources. Therefore, our project seeks to extend this work by integrating FL and TL techniques to enable collaborative model training without data sharing. To sum up, this paper establishes a robust technical foundation for segmentation accuracy and model architecture, whereas PrivaFed tackles the subsequent task of ensuring that these solutions are secure, decentralized, and in line with real-world healthcare data privacy regulations.

3.2.3 Deep Learning Models for Ischemic Stroke Lesion Segmentation in Medical Images: A Survey

3.2.3.1 Summary of the research item

Luo et al. provide an in-depth overview of DL based ischemic stroke lesion segmentation in both MRI and Computed Tomography (CT), covering the developments of the recent five-year period from 2015 to 2024 [3]. The methods are divided into architecture type, data modality, and lesion stage (acute, subacute, and chronic). Three large publicly available datasets that are used as references for stroke lesion segmentation are:

- ATLAS v2.0: 1,271 chronic stroke T1-weighted MRI.
- ISLES: several challenges (2015-2022) of DWI, FLAIR, Apparent Diffusion Coefficient (ADC), and perfusion MRI; the 2022 challenge has more than 400 patients across 19 centers.
- AISD: 397 DWI and FLAIR scans that are annotated with acute infarct segmentation.

The survey reviews core CNN-based architectures (U-Net, V-Net, UNet++, Attention U-Net, nnU-Net) and transformer-based models (UNETR, Swin UNETR, UNETR++, MedFormer) for stroke lesion segmentation. On benchmark datasets, nnU Net and UNETR achieve top DSC above 0.80 on ISLES 2022 and 0.70 to 0.78 on ATLAS v2.0. Stroke specific models such as D UNet, SPiN, MAPPING, Factorizer, and SEAN tackle challenges in detecting small lesions and handling differences in imaging modalities. The study highlights improvements through the use of multiple imaging types including DWI, ADC, and FLAIR, along with approaches like learning from partially labeled data and modeling prediction uncertainty. It also identifies ongoing challenges such as limited availability of labeled data, variation across medical centers, and the high computational cost of transformer based methods. To address these issues, it recommends federated learning, adaptation across domains, and pretraining methods that learn from unlabeled data to build scalable and privacy preserving clinical systems.

3.2.3.2 Critical Analysis

Strengths

- Offers a complete overview of DL methods, including CNNs, transformers, and hybrid approaches.
- Provides the summaries of important datasets (ATLAS, ISLES, AISD) containing explicit information on the number of subjects, modalities, and annotations.
- Recognizes new architectures (e.g., Swin UNETR, MedFormer) and their contribution to the segmentation of ischemic lesions.
- Describes research gaps, especially regarding privacy, domain adaptation, and cross-center generalization, which are essential in real-world applications.

Weaknesses

- Lacks the quantitative meta-analysis of statistically comparing results of a model across datasets.
- Mainly concentrates on MRI data, which does not give much information about CT-based stroke segmentation methods.
- Limited exploration of transformer models in terms of computational cost and inference efficiency, which significantly limits scalability to clinical use.
- Does not analyze the mitigation of fairness or bias concerning age, scanner type, and lesion size.

3.2.3.3 Relationship to the proposed research work

This survey provides the conceptual foundation for PrivaFed by highlighting both the progress and limitations of existing DL methods for stroke lesion segmentation. The identified limitations, especially the lack of privacy preserving collaboration, limited generalization across hospitals, and the lack of hyper-acute datasets, are directly related to what PrivaFed wants to do. Building on these insights, PrivaFed can integrate FL and TL with multimodal MRI fusion (DWI, ADC, FLAIR) to improve segmentation stability across institutions without data exchange. Moreover, as the survey focuses on transformer based and adaptive CNN architectures, it can guide PrivaFed in selecting models that balance accuracy, efficiency, and privacy.

3.2.4 Improvement of Automatic Ischemic Stroke Lesion Segmentation in CT Perfusion Maps Using a Learned Deep Neural Network

3.2.4.1 Summary of the research item

Soltanpour et al. propose an improved deep learning model to detect ischemic stroke lesions using computed tomography perfusion (CTP) images, which is known as a valuable modality for the diagnosis of acute stroke [4]. The research presents MultiRes U-Net, a modified version of U-Net intended to manage discrepancies in lesion scale, morphology, and presentation. Further inputs consist of contralateral brain images (mirroring the unaffected hemisphere) and threshold Time to Maximum (Tmax) heatmaps, which augment the model's capacity to differentiate ischemic tissue from normal tissue. Experiments on the ISLES 2018 dataset, of 103 patients, used MRI-based ground truth. MultiRes U-Net achieved a DSC of 68.74%, Jaccard of 57.13, and mean absolute volume error of 22.62 ml, outperforming previous U-Net and thresholding methods (typical Dice 50%). The results show that combining perfusion features with a multi-scale convolutional design improves segmentation accuracy and robustness in acute ischemic stroke detection.

3.2.4.2 Critical Analysis

Strengths

- Presents MultiRes U-Net, A modified architecture that is useful in addressing multi-scale and irregular lesion segmentation.
- Unconventional input augmentation with contra-lateral brain maps and Tmax heatmaps, which improve the accuracy of localization.
- Validated on ISLES 2018, a well-established public benchmark for CTP-based ischemic stroke segmentation.

Weaknesses

- The sample size (103 patients) is relatively low, which constrains the extrapolation of the research to a wide range of clinical populations.
- The model's computational cost and inference speed are not reported, leaving real-time clinical feasibility uncertain.
- Absence of privacy-preserving or federated approaches, which are critical for multi-center deployment.

3.2.4.3 Relationship to the proposed research work

This research can offer practical lessons to PrivaFed, particularly in strategies for enriching inputs and enhancing architectural resilience. Contra-lateral mapping and perfusion-based heatmap integration would provide insights into feature engineering in PrivaFed to better localize lesions in federal conditions. Moreover, the ability of the MultiRes U-Net to generalize to the heterogeneous scales of lesions is consistent with the aim of PrivaFed to produce consistent segmentation across different datasets of hospitals. PrivaFed can close the gap between high-accuracy models and deployable clinical systems by extending the method given in this study with federated training and multimodal MRI fusion to get higher generalizability and still maintain data privacy.

3.2.5 Automated Deep U-Net Model for Ischemic Stroke Lesion Segmentation in the Sub-Acute Phase

3.2.5.1 Summary of the research item

Ruthra and Bevi present an automated DL system for sub-acute stroke lesion segmentation in FLAIR MRI using a novel multi-path residual U-Net architecture [5]. The model employs six parallel convolutional pathways per block and residual connections to preserve fine spatial detail and mitigate intensity inhomogeneity via N4 bias field correction. They trained and evaluated the method on the ISLES 2015 Sub-acute Ischemic Stroke Segmentation (SISS) dataset (28 patients) with rigorous patient-level partitioning (18 training, 5 validation, and 5 testing). The proposed network achieved a DSC of 0.85 ± 0.12 , sensitivity of 0.82 ± 0.15 , and specificity of 0.95 ± 0.04 , outperforming DRANet, 2D CNN, and Attention U-Net, and performing comparably to CSNet. It was efficient with an inference time of 45 ms per slice and demonstrated robustness as supported by statistical confidence intervals.

3.2.5.2 Critical Analysis

Strengths

- Multi-path residual design facilitates feature extraction of objects at multiple scales, which enhances the ability of the design to segment small and irregular lesions.
- Rigorous evaluation using patient-level partitioning and bootstrap confidence intervals ensures strong methodological reliability.
- Achieves competitive DSC of 0.85 with high specificity of 0.95, outperforming several state-of-the-art architectures.
- Offers open-source reproducibility with planned code and model release.

Weaknesses

- Evaluation limited to a single dataset (ISLES 2015), which constrains generalizability to multi-center or multi-protocol data.
- Small dataset size (28 patients) may not capture diverse lesion morphologies.
- Requires further multi-site validation and clinical trials before deployment.
- This model was trained exclusively with FLAIR MRI and not multimodally with DWI and ADC, which might enhance performance.

3.2.5.3 Relationship to the proposed research work

This research provides valuable insights for PrivaFed on implementing effective and precise stroke segmentation within federated systems. It is demonstrated by the multi-path residual U-Net that innovations in network architecture can enhance the precision of lesion boundaries and improve the detection of small lesions, which can be incorporated into PrivaFed's design. Furthermore, the importance of patient-level analysis for ensuring real-world robustness is highlighted by the extensive validation methodology. Although this solution is based on centralized data, PrivaFed can extend its approach by applying similar architectural concepts within a federated framework that preserves privacy and integrates multimodal MRI fusion (FLAIR, DWI, ADC) to achieve scalable and generalizable segmentation across hospitals.

3.2.6 Acute ischemic stroke lesion segmentation in non-contrast CT images using 3D convolutional neural networks

3.2.6.1 Summary of the research item

The study by Dobshik et al. developed a method to segment acute ischemic stroke lesions on Non-Contrast CT (NCCT) scans [6]. The model processes the scans in 3D instead of analyzing one slice at a time. This provides the opportunity to study the spread of a stroke lesion through various slices. They designed the model as a variant of 3D U-Net with two modifications: First, squeeze-and-Excitation (SE) blocks, that are similar to an attention mechanism. They train the network to be more attentive to the significant feature by ignoring less useful information. Second, residual connections create shortcuts between layers, so the model learns easier and does not forget the information as it gets deeper. To cope with large CT volumes and class imbalance, they devise a patch sampling strategy (balanced between lesion vs non-lesion regions) and powerful preprocessing (skull stripping, intensity clipping, normalization). The dataset used was provided by the International Tomography Center SB RAS consisting of 81 NCCT scans. Each scan was annotated by two expert radiologists. The team used 5-fold cross-

validation to evaluate performance. Their results showed a mean Dice = 0.628 ± 0.033 , sensitivity = 0.699, specificity = 0.9965, and precision = 0.619.

3.2.6.2 Critical Analysis

Strengths

- Targets CT modality, which is clinically common, so the method is relevant for broader stroke workflows.
- Uses architectural enhancements (SE blocks + residuals) to boost the performance of U-Net.
- Balanced patch sampling helps mitigate class imbalance.
- High specificity (very few false positives), which is important in medical settings.

Weaknesses

- Small dataset (81 scans) limits diversity in scanners, patient populations, and lesion types.
- Only the NCCT modality is catered, hence it does not generalize to MRI modalities or multimodal fusion.
- Centralized training does not consider privacy preservation.
- Moderate Dice (0.63), lower than MRI-based segmentation due to CT's limited contrast.
- No use of TL from larger or related datasets, a major factor limiting performance and generalization.

3.2.6.3 Relationship to the proposed research work

This study is beneficial to PrivaFed as it demonstrates how a compact 3D model, enhanced with SE blocks and residual connections, can segment stroke lesions from CT scans with good specificity. However, the model was trained on a small dataset from a single center, with no privacy measures or TL applied. PrivaFed aims to address these gaps by applying TL on larger stroke imaging datasets and training in a federated setup.

3.3 Detailed Literature Review on Transfer Learning and Model Generalization Strategies in Brain Stroke Segmentation

This section provides an overview of key strategies utilized by recent studies to enhance model generalization in stroke lesion segmentation while working with limited and diverse medical data. These

strategies include TL and domain adaptation, ensemble methods, and hybrid architectures.

3.3.1 DeepISLES: a clinically validated ischemic stroke segmentation model from the ISLES'22 challenge

3.3.1.1 Summary of the research item

DeepISLES is a robust stroke-lesion segmentation tool built by combining the top 3 submissions (SEALS, NVAUTO, SWAN) from the ISLES'22 challenge [7]. It is an ensemble built that merges methods like nnU-Net, SegResNet, and a Swin/Factorizer variant. The authors trained and evaluated on the ISLES'22 challenge data (250 train / 150 test, multiple modalities DWI/ADC/FLAIR) and then validated the ensemble on a large external Johns Hopkins Comprehensive Stroke Center dataset (JHCSC) (N = 1685). The ensemble achieves strong results with a median Dice = 0.82 and lesion-wise F1 = 0.86. and was even preferred by neuroradiologists in a blind rating test. The tool is released in user-friendly formats (Docker, web app, GUI, GitHub).

3.3.1.2 Critical Analysis

Strengths

- Proven on large, real clinical data: validated on 1,685 external JHCSC dataset.
- Combines complementary strengths of the top 3 teams, so it is more stable than any single method.
- Its lesion volumes lined up well with experts' measurements and with stroke severity scores, and radiologists tended to prefer the model's masks.
- It is ready to use through Docker, web, and graphical UI, which helps real deployment and reproducibility.

Weaknesses

- System was trained through centralized training only, hence it does not address privacy or data-sharing restrictions.
- Ensembles are heavier because of their high computation, which can be an issue for low-resource hospitals.
- External test was large but not fully multi-center, containing diversity.

3.3.1.3 Relationship to the proposed research work

DeepISLES sets a clear performance benchmark for PrivaFed by demonstrating high-quality ischemic stroke segmentation. It also offers practical ideas, including the use of ensembles and domain-specific TL that we can borrow. However, DeepISLES was trained centrally and does not address privacy or data-sharing constraints, which is exactly the gap PrivaFed aims to fill by implementing those modeling ideas inside a privacy-preserving, federated framework.

3.3.2 Robust Ensemble of Two Different Multimodal Approaches to Segment 3D Ischemic Stroke Segmentation Using Brain Tumor Representation Among Multiple Center Datasets

3.3.2.1 Summary of the research item

Jeong et al. propose a robust ensemble for 3D ischemic stroke lesion segmentation that combines two multimodal training strategies [8]. First, joint training on DWI and ADC, but inferring from DWI only. Second, channel-wise training that concatenates DWI and ADC as input. Authors use nnU-Net as the backbone and pretrain on BraTS-2021, a brain tumor dataset, before fine-tuning on stroke scans via TL. The method was tested across three datasets: ISLES'22 and two ASAN Medical Center (AMC) sets, AMC I and AMC II, totaling hundreds of scans, and achieved top performance in ISLES'22 with DSC = 78.69% and lesion-wise F1 = 82.46% while remaining relatively robust on the AMC datasets. The authors also released code and model weights to help with reproducibility.

3.3.2.2 Critical Analysis

Strengths

- Uses BraTS tumor pretraining to boost stroke performance when stroke data are limited.
- Combines two complementary multimodal strategies and ensembling, which improves robustness on the external dataset.
- Evaluated on multi-center data (ISLES'22, AMC I and II) with a high DSC and lesion-wise F1 score.
- Authors provide weights and code, easing reproduction and deployment.

Weaknesses

- Centrally trained only, hence it does not address privacy or data-sharing constraints that occur across hospitals.

- Ensembles and multimodal pipelines can be computationally heavy, which may limit use in low-resource clinical settings.
- Approach focuses on DWI and ADC modalities. However, performance on other MRI types or different modalities might vary.
- TL from tumors helps in better learning, but there is a risk of negative transfer; thus, much careful fine-tuning is needed.

3.3.2.3 Relationship to the proposed research work

This paper gives PrivaFed two practical takeaways. First, domain-specific pretraining on BraTS combined with ensembling is an effective way to jump-start stroke segmentation when labeled stroke data are limited. Second, training with multiple multimodal schemes (joint vs. channel-wise) improves robustness to differences in available MRI sequences. For PrivaFed, we can adopt these ideas by initializing the federated model from a BraTS pretrained checkpoint and using ensemble or adapter strategies to build a more accurate and real-world system.

3.3.3 Deep learning-based automatic segmentation of cerebral infarcts on diffusion MRI

3.3.3.1 Summary of the research item

The authors have developed and tested a 3D U-Net based DL pipeline to automatically segment acute ischemic infarcts on DWI [9]. They trained on a very large, multi-site Korean dataset comprising 10,820 annotated DWIs from 10 hospitals and ran systematic experiments by sub-sampling training size (217 to 8U-Net-based,661). Following this, external validation was performed on 3 independent datasets of differing sizes ($n = 2777, 50, 250$). Researchers evaluated domain adaptation by fine-tuning on small subsets of an external target dataset to observe how a small amount of target domain data changes generalization. The study measured model performance using DSC and AHD (Average HD).

3.3.3.2 Critical Analysis

Strengths

- Model achieved reliability and generalization by using a large, multi-site dataset.
- Extensive external validation across multiple datasets confirms robustness.
- Domain adaptation findings show that fine-tuning on small local samples (50–200) can greatly improve performance.
- Use of multiple evaluation metrics (DSC, AHD) gives a balanced performance assessment.

- Achieved high segmentation accuracy with DSC up to 0.80 and AHD as low as 3.48 mm across large multi-site and external datasets, demonstrating strong generalization and robustness.

Weaknesses

- Single-modality input, i.e., DWI alone, limits detection of subtle or small lesions.
- Limited data augmentation may have underused potential performance improvements.
- Poor performance on small, brain stem lesions, which are highly critical clinical cases.
- Catastrophic forgetting observed during domain adaptation, and no mitigation strategies were tested.
- Dataset limited to only one region, i.e., Korea, restricting ethnic and scanner diversity.

3.3.3.3 Relationship to the proposed research work

The paper demonstrates the need for multi-site heterogeneity. It shows that multi-site training (approximately 866 to 1,732 DWIs) substantially improves generalization compared to single-site models. However, collecting centrally annotated multi-site data is often impossible due to privacy concerns. This is the exact point for the development of PrivaFed as an FL framework to train a shared model without moving raw data. The paper's demonstration that fine-tuning on a small set of local target images, even 50 to 200, can make a small trained model perform like a huge trained model, maps directly to FL strategies, i.e., enable local fine-tuning (personalization) after global federated aggregation. Moreover, the paper demonstrates that differences in MRI vendor, field strength, slice thickness, infarct sizes, and Last-known-well-to-scan times all caused performance variability, so a model must account for non-IID data across clients as proposed by PrivaFed.

3.3.4 Neuro-TransUNet: Segmentation of stroke lesion in MRI using transformers

3.3.4.1 Summary of the research item

The paper proposes Neuro-TransUNet, a hybrid DL model combining U-Net's spatial feature extraction with SwinUNETR's transformer-based global contextual understanding for automatic stroke lesion segmentation in 3D MRI [10]. The model is trained on the ATLAS v2.0 dataset (1271 T1-weighted MRIs from 44 cohorts and 11 countries) and includes a robust preprocessing pipeline comprising resampling, N4 bias correction, normalization, and MONAI-based augmentation to standardize input data. Following training on 655 MRI volumes, Neuro-TransUNet achieved a DSC of 0.730, outperforming U-Net (0.598), HCSNet (0.697), and SQMLP-Net (0.709). It also recorded an HD95 of 18.70 and an ASSD of 1.269, reflecting improved boundary accuracy. With over 100 million trainable parameters, the model

processes full 3D volumes, capturing both local and global lesion features. It shows recognizable segmentation performance for small, large, and multiple lesions, setting a new standard for transformer based stroke lesion segmentation.

3.3.4.2 Critical Analysis

Strengths

- U-Net and SwinUNETR based hybrid design effectively combines local spatial and global contextual features.
- Comprehensive preprocessing based on resampling, bias correction, normalization, and augmentation increased robustness.
- High quantitative performance: $DSC = 0.730$, $HD95 = 18.70$, $ASSD = 1.269$, which is better than that of state-of-the-art 2D/3D models.
- 3D volumetric processing captures full anatomical context and reduces slice level inconsistencies.

Weaknesses

- The high computational cost of Neuro-TransUNet, containing approximately 100 million trainable parameters, limits clinical deployment.
- Model uses Single modality input, i.e., T1 MRI, and thus lacks multimodal fusion.
- Does not address privacy concerns or differences in data across hospitals, as the CNN transformer model is trained in a centralized setting.
- Evaluation is limited because the model is tested only on ATLAS v2.0 and not across multiple datasets.

3.3.4.3 Relationship to the proposed research work

The Neuro-TransUNet study provides strong evidence that hybrid CNN–Transformer architectures yield high segmentation accuracy for stroke lesions. However, the centralized training of the model requires aggregating sensitive MRI data into a single repository, which poses a major privacy risk that PrivaFed directly addresses. PrivaFed builds upon this by embedding such advanced architectures into an FL framework, enabling collaboration between multiple institutions without them having to share raw patient data. Furthermore, while Neuro-TransUNet showed strong results on a single dataset, PrivaFed’s integration of FL and TL will scale its generalization across heterogeneous data. PrivaFed’s goal is to convert the strengths of Neuro-TransUNet’s architecture into a distributed and privacy preserving

framework, improving both segmentation performance and clinical feasibility.

3.3.5 Enhanced Ischemic Stroke Lesion Segmentation in MRI Using Attention U-Net with Generalized Dice Focal Loss

3.3.5.1 Summary of the research item

This paper presents an improved version of the U-Net model called Attention U-Net with Generalized Dice Focal Loss (GDFL) for segmenting ischemic stroke lesions in MRI scans [11]. The goal of the study is to deal with the common problem of class imbalance, where healthy brain tissue heavily outweighs lesion pixels. The model adds an attention gate to the regular U-Net to help the network focus on important lesion regions and uses a combination of GDFL as the Loss function to handle uneven lesion sizes. The model was trained and tested on multi-site MRI scans obtained from ISLES 2015 and ISLES 2022 datasets. Preprocessing steps such as resampling, NIfTI-to-PNG conversion, normalization, and data augmentation were applied to make the data more consistent. The number of parameters is approximately 7.94 million, and the model was trained on GPUs. Results show that on ISLES 2015, the model achieved an intersection over union (IoU) of 0.8596, an F1-score of 0.9129, and an HD of 4.09, while on ISLES 2022, it reached an IoU of 0.6961, an F1 of 0.7517, and an HD of 9.90. The addition of the attention mechanism improved IoU by about 3% and reduced HD by about 1 mm, showing that the enhancements help the model detect stroke lesions more accurately.

3.3.5.2 Critical Analysis

Strengths

- The model is lightweight with around 7.9 million parameters, which makes it much easier to train and deploy compared to other large segmentation networks.
- The use of GDFL effectively handles class imbalance and improves performance on small lesion regions.
- The use of attention gates guides the network to focus on relevant areas, which provides improved segmentation accuracy.
- A comprehensive preprocessing and augmentation pipeline improves generalization across different MRI orientations.
- The model shows strong results on ISLES 2015 as it achieved one of the highest IoU and F1-scores compared to baseline U-Net models.

Weaknesses

- Drop in model's performance on ISLES 2022 dataset shows difficulty in handling multi-site and high-variance MRI data.
- The model fails to detect tiny lesions, resulting in missed detections in critical cases.
- Because the model is trained on 2D slices rather than full 3D volumes, the model can lose important spatial context between slices.
- The loss weighting values had to be adjusted for each dataset, meaning extra fine-tuning is needed before deployment in new environments, implying reduced generalization.

3.3.5.3 Relationship to the proposed research work

This research closely relates to the goals of PrivaFed, as it demonstrates how an efficient and attention-based architecture can achieve strong results while remaining computationally manageable. The Attention U-Net's lightweight design makes it suitable for FL environments where hospitals or institutions have limited computing power. Its use of GDFL also highlights how combining loss functions can help with data imbalance, an issue that appears when using different hospital datasets in a federated environment. However, the paper trains its model in a centralized setting, which raises privacy concerns. PrivaFed aims to overcome this by training similar or more advanced architectures, such as Swin Transformers, in a federated and privacy-preserving manner. PrivaFed will therefore extend the strengths of this approach through the provision of data privacy and cross-site adaptability.

3.4 Detailed Literature Review on Federated Learning in Brain Stroke Segmentation

This section examines the use of FL frameworks in medical imaging, particularly for stroke lesion segmentation. It explains how distributed training keeps data private, makes it easier for institutions to work together, and helps create stronger and more general models.

3.4.1 Federative ischemic stroke segmentation as an alternative to overcome domain-shift multi-institution challenges

3.4.1.1 Summary of the research item

Rangel and Martinez introduce an FL model for ischemic stroke lesion segmentation in multi-institutional MRI data [12]. The approach addresses the challenges of domain shift and data privacy by enabling joint training across 14 simulated clinical centers without exchanging patient information. The study is based on nnU-Net and utilizes more than 2,000 publicly available DWI and ADC scans (ISLES'22,

ISLES'24, SOOP) along with private datasets from Colombian academic institutions (FOSCAL and FOSUNAB). Several federation aggregation rules (FedAvg, VanillaAvg, Beta Weighting, Softmax, and FedProx) were compared with a centralized control. FedAvg achieved the best overall performance ($DSC = 0.71 \pm 0.24$, $Local F1 = 0.70 \pm 0.26$); it not only outperformed the centralized and other federated schemes but also maintained a similar level of accuracy when trained on large datasets and evaluated on limited test-only centers. The model demonstrated strong generalization to unseen data, achieving $DSC = 0.64 \pm 0.29$ in out-of-distribution institutions without retraining.

3.4.1.2 Critical Analysis

Strengths

- Effectively demonstrates FL as a privacy-preserving alternative for multi-institution stroke segmentation.
- Provides freedom of choice of open-source code and reproducible experimental setup, promoting transparency and reuse.
- Widespread validation across 14 heterogeneous centers, including both public and privately available datasets of different scanner vendors and lesion morphology.
- Provides detailed analysis per lesion volume (small, medium, large) and includes both overlap (DSC , $Local F1$) and clinical (Absolute Volume Difference (AVD), Average Lesion Difference (ALD)) metrics.

Weaknesses

- Relies solely on DWI and ADC modalities; multimodal fusion with FLAIR or CT could further enhance robustness
- Requires considerable computational infrastructure at participating centers for local training.
- Lacks domain adaptation or personalized FL strategies, which might enhance the performance of local centers.

3.4.1.3 Relationship to the proposed research work

The research directly influences the design philosophy of PrivaFed by demonstrating that FL is effective for privacy-preserving cross-institution stroke segmentation. It emphasizes that FedAvg can deliver comparable or even better performance than centralized models while ensuring data privacy. Additionally, the incorporation of DWI and ADC modalities, along with validation across diverse datasets, showcases the effectiveness of federated training in enhancing generalization, a concept essential to PrivaFed's goal

of deploying models across hospitals with varying patient populations. Future versions of PrivaFed can incorporate adaptive and aggregation strategies, as proposed by Rangel and Martinez, to further enhance scalability and efficiency in practical federated settings.

3.4.2 In-silo federated learning vs centralized learning for segmenting acute and chronic ischemic brain lesions

3.4.2.1 Summary of the Research Item

Kim et al. evaluated the effectiveness of FL versus centralized learning for segmenting acute infarct and chronic white matter hyperintensity (WMH) lesions in brain MRI scans [13]. The study used a large multicenter dataset of 13,546 DWI images from 10 hospitals and 8,421 FLAIR images from 9 hospitals for Tasks I and II, respectively. A central server aggregated the models, updating every four rounds, using FedYogi for Task I and FedAvg for Task II, with a batch clipping strategy to handle data heterogeneity and reduce idle time.

In Task I, the FL model with batch clipping (360 epochs) achieved a Dice score of 0.754 ± 0.183 , outperforming the centralized learning model (0.691 ± 0.229 ; $p = 0.001$) and matching the best centralized learning model (0.755 ± 0.207) trained for 940 epochs. With 1000 epochs, FL further improved to 0.776 ± 0.170 . In Task II, all models showed similar performance ($DSC \approx 0.75$). Batch clipping halved training time from 3.5 to 1.75 hours. Overall, the study confirms FL's viability for medical image segmentation across institutions by demonstrating that it can match centralized learning performance while maintaining data privacy.

3.4.2.2 Critical Analysis

Strengths

- The study utilizes a large, diverse set of datasets (over 20,000 MRI scans), enhancing the external validity of the findings.
- An extensive comparison of FL and centralized learning under the same experimental conditions.
- Introduction of batch clipping improves efficiency and reduces idle time among clients.
- Demonstrates the feasibility of privacy-preserving FL without significant accuracy loss.
- Strong statistical validation, including significance testing (paired t-tests, ANOVA).
- Ensures reproducibility of performance with a well-documented methodology and open-access framework.

Weaknesses

- Tests are limited to only two MRI modalities (DWI and FLAIR), with no exploration of other imaging types.
- Requires high computational infrastructure, which may not be available in smaller clinics.
- Batch clipping, while effective, does not comply with theoretical FL aggregation assumptions.
- One-shot aggregation and few-shot FL performed poorly, indicating that FL continues to rely on regular communication.

3.4.2.3 Relationship to the Proposed Research Work

The paper is highly applicable to PrivaFed as it shows that FL can achieve the performance of centralized segmentation while keeping patient data secure across institutions. The integration of batch clipping provides a practical approach to optimizing communication and computation in federated setups, serving as a useful mechanism for scaling PrivaFed to heterogeneous clinical systems. Moreover, the FedAvg and FedYogi aggregation schemes can offer insights into the selection of adaptive optimizers for multi-center FL. Overall, the findings support the feasibility of decentralized medical AI that preserves sensitive information, aligning with the privacy objectives of PrivaFed.

3.4.3 FedUniBrain: Feasibility of Federated Learning from Client Databases with Different Brain Diseases and MRI Modalities

3.4.3.1 Summary of the research item

Wagner et al. introduce FedUniBrain, a FL approach that trains a single 3D Residual U-Net model across multiple brain MRI databases, even when each hospital has different diseases and MRI scan types [14]. They combined seven public datasets (including BRATS, ATLAS, and ISLES), covering tumors, stroke, MS, TBI, and ischemic lesions. To handle differences between sites, they trained the model with a union of all MRI modalities (T1, T1c, FLAIR, T2, PD, SWI, DWI) and used a simple trick called modality drop, where some scans are randomly hidden during training. This made the model robust when certain scans were missing at test time. They also tested different normalization strategies, including BatchNorm (BN), client-specific BN (FedBN), InstanceNorm (IN), and Normalization-Free (NF). Results showed the federated model performed with an average DSC of around 0.691, and it generalized well to new hospitals and unseen modality combinations.

3.4.3.2 Critical Analysis

Strengths

- It performs effectively on a variety of datasets and medical conditions.
- Techniques such as modality drop and normalization tailored to individual clients are straightforward methods that lead to significant enhancements.
- Demonstrates that federated models have the ability to generalize to unseen data in new hospitals.
- Proves to be robust when MRI scans are missing at test time.

Weaknesses

- Does not evaluate privacy-preserving primitives such as differential privacy, secure aggregation, or attack resilience.
- Uses only one model (U-Net), so results may not generalize to other models.
- FedBN gave better accuracy for the hospitals involved in training, but it reduced generalization to new hospitals. In contrast, IN or NF methods generalized better to unseen clients, though with slightly weaker results on the original ones.
- Paper does not measure real communication overhead, latency, or compute constraints on small clinical sites, which is important in practice.
- Still struggles when facing diseases or scan types never seen before.

3.4.3.3 Relationship to the proposed research work

The FedUniBrain study supports the motivation for our project, as it demonstrates that FL can successfully train a single model across hospitals with different brain diseases and MRI modalities. The use of union modalities with modality drop and normalization strategies in the study provides useful insights for handling missing scans and dataset variations, which is closely linked with stroke lesion segmentation. However, the main focus of the study is on model feasibility and performance, without addressing the issue of privacy protection or efficient deployment, which are central to PrivaFed. Building on their findings, our project integrates privacy preserving techniques and TL to develop a secure framework that not only achieves high segmentation accuracy but also ensures patient confidentiality and real world practicality.

3.4.4 Prediction of Brain Stroke Using Federated Learning

3.4.4.1 Summary of the research item

The study by Tehseen et al. focuses on predicting brain stroke using FL to protect patient privacy [15]. The authors worked with the Kaggle Stroke Prediction Dataset, which contains 5,110 patient records including age, gender, glucose level, BMI, hypertension, and smoking status. They trained several models, including Logistic Regression, Decision Tree, and K-Nearest Neighbors, and later combined them with a Weighted Voting classifier. Random Forest provided the best results among these model. It achieved about 97% accuracy, 95% precision, and an F1 score of 97%. Although the study addresses a classification problem instead of segmentation, it shows that FL can deliver strong prediction performance while keeping patient data private, making it promising for real healthcare applications.

3.4.4.2 Critical Analysis

Strengths

- Ensures data privacy by allowing hospitals to train locally without sharing raw information.
- Achieved very high accuracy of 97%.
- A good variety of patient features was used which helped in improving the reliability of the model.
- Showed that FL can be applied even with limited resources and non-image data.
- Used feature selection and SMOTE as the preprocessing steps which helped in balancing the dataset and improving learning.

Weaknesses

- The study only used tabular (numerical) data and did not cater medical imaging.
- The FL setup was not tested across real hospitals or systems because it was tested in a simulated environment.
- The approach did not use TL, which could have helped reuse knowledge from other datasets.
- The main goal of the study was stroke prediction rather than lesion segmentation or visual localization.

3.4.4.3 Relationship to the proposed research work

This study is relevant to PrivaFed as it shows that FL can achieve high accuracy without sharing the private medical data of the patients. However, it focused only on tabular data and treated stroke pre-

diction as a simple classification task. PrivaFed builds on this idea by moving from classification to image-based stroke segmentation, using TL and a privacy-preserving federated setup to handle diverse MRI data across hospitals.

3.5 Literature Review Summary Table

This section presents summary tables of the reviewed literature, organized into three main categories: Brain Stroke Segmentation, TL and Model Generalization Strategies in Brain Stroke Segmentation, and FL in Brain Stroke Segmentation. Table 3.1, Table 3.2 and Table 3.3 highlight key findings, methodologies, results, and limitations of prior research, providing a concise comparative overview of related work in privacy-preserving stroke lesion segmentation and medical image analysis.

3.5.1 Brain Stroke Segmentation

Table 3.1: Brain Stroke Segmentation Literature Review Summary Table

Author	Method	Results	Limitations
Gheibi et al. [1]	Residual CNN combining U-Net and ResNet blocks; multimodal MRI input (DWI + FLAIR); trained on local dataset and SPES 2015.	Achieved DSC 85.43% (local dataset) and 79.23% (SPES 2015); faster training with 6.4M parameters.	Small dataset (44 scans); 2D model limits spatial context; no federated setup; weak small-lesion detection; centralized training.
Foulon et al. [2]	Proposed a transformer-based segmentation model (Swin-UNETR, Swin-UNETR+Ctr) for ischaemic stroke lesion segmentation in DWI MRI. Trained on a multi-site internal dataset (3,563 lesions, 6,900 controls) using Thresholded Average loss and 5-fold cross-validation.	Swin-UNETR: DSC 0.8952, HD 2.65 voxels; Swin-UNETR+Ctr: DSC 0.8915, HD 2.66; both outperform U-Net (DSC 0.84, HD 3.92) with strong generalization.	Centralized training raises privacy issues; no labeled external test limits validation; computationally heavy; struggles on small infarcts; lacks federated setup for distributed use.
Luo et al. [3]	Survey of deep learning architectures (CNN, Transformer, hybrid) for ischemic stroke lesion segmentation. Reviews public datasets (ATLAS, ISLES, AISD), models, loss functions, and challenges.	Dice scores quoted only when reported by authors. Factorizer 0.694 (HD95 6.8 mm) ISLES-2022; SAN-Net 0.701 ATLAS-R1.2; SEAN 0.589 AISD; Boundary+Dice $\approx +13\%$ ISLES-2015; ICI +1.7-3.7% ATLAS; Swin-UNETR 0.64 overall ATLAS-R2.0 (≈ 0.35 for <5 mL lesions, ≈ 0.80 for >30 mL).	Lacks experimental validation; focuses on secondary data; does not benchmark models quantitatively.
Soltanpour et al. [4]	MultiRes U-Net for CT perfusion maps; uses contra-lateral and T _{max} features.	Dice $\approx 68\%$, Jaccard $\approx 57.13\%$ (ISLES 2018).	High volume error (≈ 22.6 ml); moderate accuracy for small lesions
Ruthra & Bevi [5]	Developed a multi-path residual U-Net with six parallel pathways per block (2.34M parameters). Used 5-fold cross-validation, N4 bias correction, and bootstrap CI for robust validation.	Achieved DSC = 0.85 ± 0.12 (val) and 0.89 ± 0.07 (test), specificity = 0.95 ± 0.04 , sensitivity = 0.82 ± 0.15 , Hausdorff distance = 14.1 ± 5.8 mm. Outperformed DRANet, 2D CNN, and Attention U-Net ($p < 0.01$).	Limited intensity-robustness, small-lesion detail loss, data-leakage-prone validation, and missing runtime metrics still hinder clinical deployment.
Dobshik et al. [6]	Modified 3D U-Net with squeeze-and-excitation blocks and residual connections for 3D scans. Used balanced patch sampling and preprocessing on 81 NCCT scans from the International Tomography Center SB RAS.	Achieved mean Dice $\approx 0.63 \pm 0.03$, sensitivity ≈ 0.70 , specificity ≈ 0.996 , precision ≈ 0.62 (5-fold cross-validation).	Small, single-center dataset; only NCCT modality; centralized training (no FL/privacy).

3.5.2 Transfer Learning and Model Generalization Strategies in Brain Stroke Segmentation

Table 3.2: Transfer Learning and Model Generalization Strategies in Brain Stroke Segmentation Literature Review Summary Table

Author	Method	Results	Limitations
de la Rosa et al. [7]	Post challenge ensemble built from top ISLES'22 submissions. Reused and fine tuned top models; trained on ISLES'22 and validated on a large external Johns Hopkins cohort.	Median Dice ≈ 0.82 , lesion-wise F1 ≈ 0.86 . In a blind test, neuroradiologists often preferred model masks.	Centralized training only. Ensembles are computationally heavier for training and inference. External validation is large but could be even more diversely representative.
Jeong et al. [8]	Two multimodal strategies ensembled using nnU-Net: (1) joint DWI+ADC training with inference from DWI-only, (2) channel-wise DWI+ADC concatenation. Uses BraTS (brain tumor) pretraining for transfer learning then fine-tunes on stroke data; tested on ISLES'22 and AMC I & II.	Dice $\approx 78.7\%$, lesion-wise F1 $\approx 82.5\%$. Shows improved robustness across the AMC external sets and benefits from BraTS pretraining when stroke labels are limited.	Centralized training only. Ensembles are computationally heavy. Risk of negative transfer if pretraining domain differs too much from target.
Ryu et al. [9]	3D U-Net trained on 10,820 DWI scans using multi-site data; domain adaptation and external validation across datasets.	Achieved maximum DSC $\approx 0.80 \pm 0.13$ and AHD ≈ 3.48 mm on multi-site dataset; DSC ≈ 0.73 – 0.74 on external datasets after fine-tuning.	Single-modality (DWI) only; poor small-lesion segmentation; Catastrophic forgetting during domain adaptation; centralized setup; region-specific dataset.
Nouman et al. [10]	Hybrid CNN–Transformer model (U-Net + SwinUNETR) trained on ATLAS v2.0; 3D volumetric segmentation with MONAI preprocessing.	Achieved DSC 0.730, HD95 18.70, ASSD 1.269; outperformed U-Net (0.598) and HCSNet (0.697).	High computational demand (approx 100M params); single modality input; centralized training; limited external validation.
Garcia-Salgado et al. [11]	2D Attention U-Net trained with combined Generalized Dice and Focal Loss on ISLES 2015 and ISLES 2022 datasets.	On ISLES 2015: IoU 0.8596, F1 0.9129, HD 4.09; On ISLES 2022: IoU 0.6961, F1 0.7517, HD 9.90.	2D slice-based approach; requires dataset-specific tuning; reduced accuracy on small lesions; centralized setup.

3.5.3 Federated Learning in Brain Stroke Segmentation

Table 3.3: Federated Learning in Brain Stroke Segmentation Literature Review Summary Table

Author	Method	Results	Limitations
Rangel & Martínez [12]	Developed Federated Learning (FedAvg) using 14 simulated healthcare centers and 2,031 DWI/ADC MRI studies for ischemic stroke segmentation. Compared 5 aggregation algorithms.	FedAvg achieved DSC = 0.71 ± 0.24 , outperforming centralized models and showing generalization on unseen centers (DSC = 0.64 ± 0.29).	Based on emulated rather than real hospitals; heterogeneous data quality; small lesions remain difficult to segment.
Kim et al. [13]	Experimental comparison of Federated Learning (FedYogi, FedAvg) and Centralized Learning (CL) for acute infarct (Task I, DWI; 13,546 scans from 10 hospitals) and white matter hyperintensity (Task II, FLAIR; 8,421 scans from 9 hospitals) segmentation.	FL achieved Dice = 0.754 ± 0.183 (acute) and 0.761 ± 0.299 (chronic), comparable or superior to CL ($p < 0.001$). Batch clipping reduced training time by 50%.	Limited to Non-IID. institutional data; few-shot FL underperformed; lacks validation on multi-modal imaging.
Wagner et al. [14]	Federated training of a single 3D Residual U-Net across 7 public brain MRI datasets (different diseases & modality sets). Uses union-of-modalities input, random modality-drop during training, and experiments with normalization schemes (FedBN, IN, GN, NF).	FL achieved best average Dice $\approx 69.1\%$. Modality-drop made the model robust to missing MRI sequences. Model generalized well to unseen clients and unseen modality combinations.	Uses one model (U-Net) so architecture generality is untested, and it lacks measurements of communication cost or attack resilience which is all important for real hospital deployment.
Tehseen et al. [15]	FL setup using multiple ML classifiers (Logistic Regression, Decision Tree, KNN, Random Forest) on the Kaggle Stroke Prediction Dataset. Combined outputs using a weighted voting approach.	Best performance achieved by Random Forest with 97% accuracy, 95% precision, and 97% F1-score.	Used tabular data instead of MRI images; simulated FL setup (not real hospitals); no TL applied.

3.6 Conclusion

The literature review indicates that there has been considerable advancement in the brain stroke lesion segmentation using the DL, FL, and TL. Transformer-based and state of the art architectures are very accurate in segmentation. However, the majority of approaches were based on centralized data and it is of significant concern due privacy and model generalization. The FL studies showed good performance across heterogeneous datasets because of model collaboration without sharing the data. Additionally, TL and model generalization strategies improves performance in limited data. However, challenges such as small-lesion detection, variability in modality, and computational complexity remain open areas for improvement. These areas have influenced the incentive of designing privacy-preserving, scalable, and clinically robust segmentation models such as PrivaFed.

Chapter 4 Software Requirement Specifications

The Software Requirement Specifications (SRS) document provides a complete description of the project. It specifies the functional, non-functional, hardware, and software requirements. Additionally, it includes detailed use cases and graphical user interfaces for our users who will be medical practitioners.

4.1 List of Features

The proposed system will provide the following features:

- Pre-processing of brain scans.
- Enabling multiple clients to collaboratively train a model without sharing raw brain scan data.
- Building optimized models to reduce communicational cost and computational load.
- Incorporating pre-trained models to improve performance.
- Addressing heterogeneity by adapting global models to client-specific data.
- Providing a web-based interface allowing users to perform different functionalities.
- Uploading MRI brain scans for analysis.
- Displaying the detected stroke lesion regions on the uploaded scan.
- Generating structured reports based on the segmentation results, providing clinicians with a second opinion.

4.2 Functional Requirements

The following functional requirements shall be available in the system.

1. **User Authentication:** The system shall verify the identity of users before granting access.
2. **Upload Scan:** The system shall allow the user to upload brain MRI scans.
3. **Stroke Lesion Segmentation:** The system shall be able to correctly segment stroke lesions in the MRI scan.
4. **Result Visualization:** The system shall visually display the segmented brain lesions on the scan.
5. **Report Generation:** The system shall generate a report summarizing the segmented result.
6. **Federated Model Training:** The system shall train local models using the data of each client institution, and the resulting updates shall be transmitted to the central server for aggregation.

7. **Model Personalization:** The system shall refine the global model using local client data to enhance accuracy for each site.

4.3 Quality Attributes

The system is intended to show the following quality attributes:

- **Reliability:** The system shall show guaranteed performance in model training, segmentation, and report creation without unexpected system failures.
- **Security:** The system will safeguard sensitive medical information by limiting access to authorized users and prohibiting the storing of raw brain scans in the central database.
- **Usability:** The system shall provide medical professionals with a user friendly interface that has capabilities for report generation, stroke lesion visualization, and uploading the scans.
- **Maintainability:** The system will be designed with a modular architecture to enable future upgrades, debugging, and additions.
- **Efficiency:** The system shall utilize lightweight models and optimized updates to minimize computational and communication costs.

4.4 Non-Functional Requirements

The non-functional requirements for PrivaFed are as follows:

1. **Performance:** The system should allow the user to interact with the web application in 2 to 5 seconds in order to allow for real time clinical use.
2. **Reusability:** By reusing model components and the training workflow on stroke imaging datasets, the approach should enable retraining with minor adjustments.
3. **Availability:** The system should remain running and operational with minimum downtime during inference, while gracefully recovering from any unforeseen failures.
4. **Privacy Compliance:** The system should ensure that no patient data is shared between clients or unauthorized personnel, as required by practices concerning health care data privacy.
5. **Robustness:** The system should be able to work with large, disparate datasets without having to experience data loss or system failure.

4.5 Assumptions

The assumptions made for the specification for the user to use the system are listed as:

- The user is a licensed medical practitioner.
- The user has a basic understanding of the purpose of the application.
- The user possesses sufficient knowledge of how to use a computer and a website.
- The user provides good-quality MRI scans in the correct format.
- The user does not make any malicious attempt to misuse the system.

4.6 Use Cases

Given below are the use cases identified in our web application.

Table 4.1: Sign Up/ Sign In

Name		Sign Up/ Sign In	
Actors		User	
Summary		User signs up and signs in to the system using a Magic Link sent to their email, instead of entering a password.	
Pre-Conditions		User must provide a valid and accessible email address. The system must be able to send emails through Magic Link’s authentication service.	
Post-Conditions		For Sign Up: A new account is created if the email is not already registered. For Sign In: User is logged into the application.	
Special Requirements		1- Valid email address. 2- Secure, one-time-use link with expiration time. 3- Internet connectivity for email verification.	
Basic Flow			
Actor Action		System Response	
1	User selects Sign Up / Sign In option.	2	System displays an email input form.
3	User enters their email address.	4	System requests Magic Link service to generate a secure authentication link and sends it to the provided email.
5	User opens their email and clicks the Magic Link.	6	System verifies the validity of the link. If valid, user is redirected to the dashboard
Alternative Flow			
3	User enters an invalid email.	3-A	System displays the message: ‘Invalid email address.’
5	User does not receive the email due to delivery issues.	5-A	System prompts: “Resend Magic Link”
6	User clicks an expired or already used Magic Link.	6-A	System displays: “Link expired or invalid. Please request a new link.”

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Table 4.2: Logout

Name		Logout	
Actors		User	
Summary		User can securely logout of the system, ending their current session so no unauthorized access occurs once they leave the application.	
Pre-Conditions		User must be logged into the system.	
Post-Conditions		Current user session is terminated and user is redirected to login page.	
Special Requirements		Session tokens must be expired to prevent reuse.	
Basic Flow			
Actor Action		System Response	
1	User selects 'Logout' option.	2	System terminates the user session and clears all session tokens.
		3	User is redirected to login page.
Alternative Flow			
1	System fails to terminate the session due to a server issue.	2-A	System displays the message: 'Failed to logout. Please try again.'

Table 4.3: Upload Brain Scan

Name		Upload Brain Scan	
Actors		User	
Summary		User uploads brain scan images into the PrivaFed system. The system then prepares the scans for analysis.	
Pre-Conditions		User is registered and logged into the system.	
		User has selected an existing patient from their registered patient list.	
		The brain scan image is in a supported format.	
Post-Conditions		The brain scan is successfully uploaded and queued for analysis.	
Special Requirements		High-resolution scan images.	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Upload Scan' option	2	System display the upload interface.
3	User uploads the brain scan image	4	System confirms the upload and queues the scan for analysis.
Alternative Flow			
3	User attempts to upload an unsupported file format.	4-A	System displays error message: 'Unsupported File Format'.

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Table 4.4: View Analysis Results

Name		View Analysis Result	
Actors		User	
Summary		Users access and review the analysis results of uploaded MRI scans.	
Pre-Conditions		User is logged into the system.	
		User has selected an existing patient from their registered patient list.	
		The brain scan has been uploaded and analyzed by the system.	
Post-Conditions		User has viewed and understood the analysis results.	
Special Requirements		None.	
Basic Flow			
Actor Action		System Response	
1	User selects the 'View Results' option for the scan.	2	System displays the analysis results.
Alternative Flow			
1	User selects a scan that is still being analyzed.	2-A	System displays the message: 'Analysis still in progress. Check again in a few minutes.'

Table 4.5: Generate Report

Name		Generate Report	
Actors		User	
Summary		After performing brain scan analysis, the system generates a detailed report containing the segmented MRI with highlighted stroke lesions and other relevant findings.	
Pre-Conditions		The brain scan image for a specific patient has been analyzed, and the results are ready to be compiled in report format.	
Post-Conditions		A detailed segmentation report is generated and is available for review or download.	
Special Requirements		The report must be clear, concise, and reliable.	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Generate Report' option for the scan.	2	System compiles the analysis results into a comprehensive report format.
3	User reviews or downloads the generated report.	4	System provides options to either review on-screen or download the report.
Alternative Flow			
3	An error occurs during report generation.	4-A	System displays error message and suggests retrying or contacting support.

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Table 4.6: Edit Profile

Name		Edit Profile	
Actors		User	
Summary		Allows an authorized user to edit their own personal records in the system.	
Pre-Conditions		User must be logged into the system through an authorized account.	
Post-Conditions		The user record is successfully linked to the user’s account.	
Special Requirements		Each user is uniquely identified by their CNIC.	
Basic Flow			
Actor Action		System Response	
1	User selects the ”Edit Profile” option.	2	System displays a form to change basic user details.
3	User make changes as desired and clicks ”Save”.	4	System validates the information.
		5	System successfully updates the profile.
		6	System displays a confirmation message “Profile updated successfully.”
Alternate Flow			
3-A	User tries to change CNIC.	3-A	System displays CNIC as an unchangeable parameter
4-A	User cancels the update.	4-A	System redirects the user to home page.

Table 4.7: Add Patient

Name		Add Patient	
Actors		User	
Summary		Allows an authorized user to add a new patient to their personal records in the system.	
Pre-Conditions		User must be logged into the system through an authorized account.	
		The patient being added should not already exist in the user’s record.	
Post-Conditions		The patient record is successfully created and linked to the user’s account.	
Special Requirements		Each patient is uniquely identified (by an internal patient ID).	
		Data privacy and compliance with healthcare regulations must be ensured.	
Basic Flow			
Actor Action		System Response	
1	User selects the "Add Patient" option.	2	System displays a form to enter basic patient details (CNIC, name, age, gender, contact number)
3	User fills out the patient details and click "Save".	4	System validates the information and checks for duplicates.
		5	System creates a new patient record and links it to the user’s profile.
		6	System displays a confirmation message "Patient added successfully."
Alternate Flow			
3-A	User enters incomplete or invalid data.	3-A	System displays "Invalid or incomplete details. Please review and try again."
4-A	Patient already exists in the user’s record.	4-A	System displays: "Patient already exists in record."

Table 4.8: Search Patient

Name		Search Patient	
Actors		User	
Summary		Allows an authorized user to search a patient(s) record in the system.	
Pre-Conditions		User must be logged into the system through an authorized account. The user is at patient dashboard page.	
Post-Conditions		The patient record is successfully created and linked to the user’s account.	
Special Requirements		Search should support partial names. System should display search results dynamically as the user types. Search results should appear instantly without requiring a page reload	
Basic Flow			
Actor Action		System Response	
1	User enters patient name in the search bar.	2	System displays matching results.
Alternate Flow			
3-A	User enters invalid patient name.	3-A	The system displays a “No records found” message .

Table 4.9: Update Patient

Name		Update Patient	
Actors		User	
Summary		Allows an authorized user to update an existing patient linked to their personal records in the system.	
Pre-Conditions		User must be logged into the system through an authorized account. The patient being edited should already exist in the user’s record.	
Post-Conditions		The patient record is successfully edited and updated.	
Special Requirements		The system shall provide an option to cancel or discard changes before saving. Data privacy and compliance with healthcare regulations must be ensured.	
Basic Flow			
Actor Action		System Response	
1	User selects a patient from dashboard.	2	System redirects to patient profile page.
3	User clicks ”Edit” button.	4	System opens a form.
5	User changes desired fields and clicks ”Save”.	6	System updates the patient profile and displays a confirmation message.
Alternate Flow			
3-A	User enters invalid data.	3-A	System displays “Invalid details. Please review and try again.”
4-A	Patient clicks ”Cancel” button.	4-A	System redirects back to patient profile page.

Table 4.10: Delete Patient

Name		Delete Patient	
Actors		User	
Summary		Allows an authorized user to delete an existing patient.	
Pre-Conditions		User must be logged into the system through an authorized account. The patient being deleted should already exist in the user’s record.	
Post-Conditions		The patient record is successfully deleted.	
Special Requirements		The system shall require user confirmation before deleting a patient record.	
Basic Flow			
Actor Action		System Response	
1	User selects a patient from dashboard.	2	System redirects to patient profile page.
3	User clicks "Delete" button.	4	System prompts a confirmation.
5	User clicks "Confirm".	6	System deletes the patient.
		7	System displays a message "Patient deleted successfully."
Alternate Flow			
3-A	User clicks "Cancel".	3-A	System aborts the deletion process and returns to previous screen.

Table 4.11: View Patient History

Name		View Patient History	
Actors		User	
Summary		Allows a user to view the historical data of a specific patient in his patient list.	
Pre-Conditions		User is registered and logged into the system. User has one or more patients associated with his account.	
Post-Conditions		User successfully views the selected patient’s complete medical history.	
Special Requirements		All historical data must be stored securely and accessed only by the authorized user.	
Basic Flow			
Actor Action		System Response	
1	User presses the “View Patient History” button.	2	System retrieves and displays all associated segmented MRI images and reports for that patient along with MRI upload dates.
Alternate Flow			
1-A	User has no patients in their record.	1-A	System displays “Record Empty.”
2-A	Selected patient has no previous record.	2-A	System displays “No historical data available for patient.”

4.7 Hardware and Software Requirements

This section discusses the hardware and software requirements needed to develop and deploy the project.

4.7.1 Hardware Requirements

The hardware requirements of the project are as follows:

- Personal computer or laptop.
- Stable internet connection.
- Sufficient local storage for datasets and model checkpoints.
- server with sufficient capacity to support model aggregation and federated training.
- GPU

4.7.2 Software Requirements

Below are the software requirements for the operation of the system:

- **Development Tools:** VS Code, Google Colaboratory, Kaggle.
- **Programming Languages:** Python, JavaScript, TypeScript.
- **Frontend Frameworks and Styling:** Next.js, Tailwind CSS, Shadcn UI.
- **Backend and Database:** Supabase.

4.8 Graphical User Interface

The following figures provide an overview of PrivaFed's user interface screens as a demonstration of its design, layout, and user interaction flow.

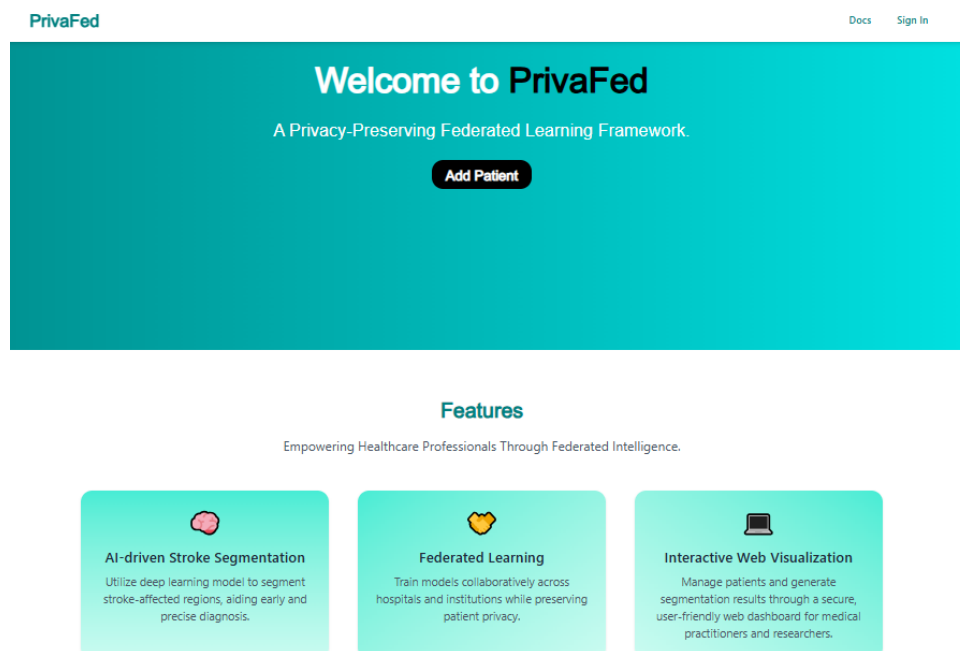


Figure 4.1: Home Page

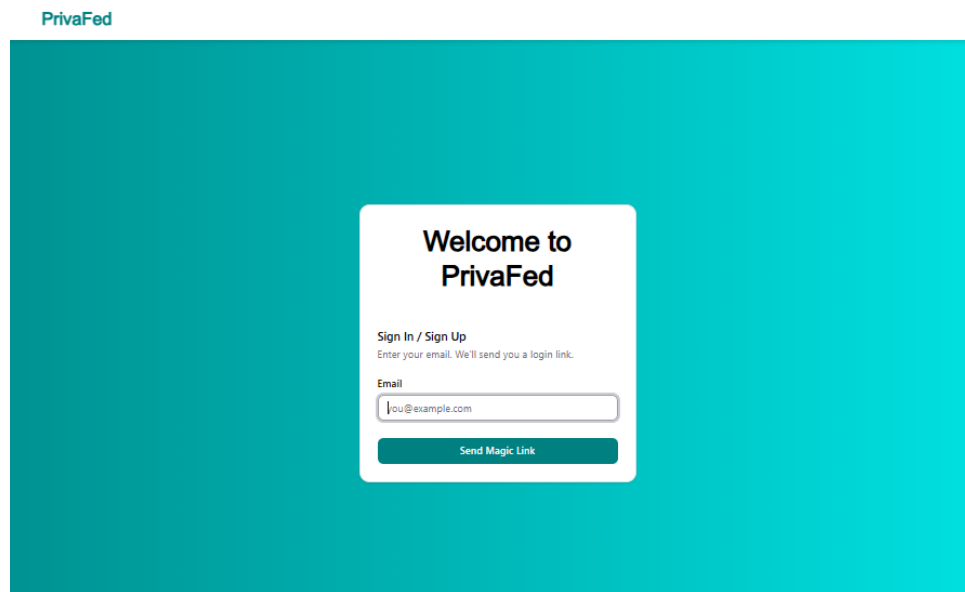


Figure 4.2: Sign Up / Sign In Page

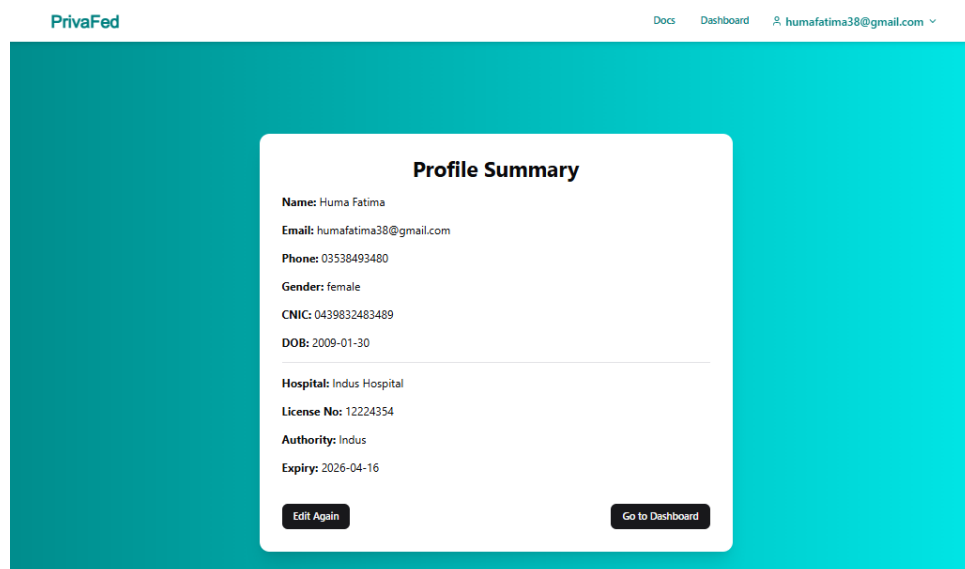


Figure 4.3: User Profile Page

Figure 4.4: Edit User Profile Page

Figure 4.5: Dashboard

Figure 4.6: Add Patient Page

The screenshot displays the 'Patient Profile' page in the PrivaFed application. The page header includes the PrivaFed logo, navigation links for 'Docs' and 'Dashboard', and a user profile for 'humafatima38@gmail.com'. The main content area is titled 'Patient Profile' and includes a '+ Add New Scan' button. Below this, the patient's CNIC is listed as '42008-2928490-2'. The 'Personal Details' section shows the patient's name as 'Ali Anwar', date of birth as '2/11/2013', gender as 'male', and age as '13 years'. It also displays the CNIC number '42008-2928490-2'. At the bottom of this section, it indicates the profile was created on 5/4/2026 at 5:40:49 AM and last updated on 5/4/2026 at 5:41:05 AM. The 'Scan History' section shows a single scan labeled 'MRI' with a thumbnail image. It includes the text '1st scan', 'Uploaded: 5/4/2026, 5:41:37 AM', and 'Analysed: 5/4/2026, 6:01:48 AM'. There are buttons for 'View Result' and 'Report' next to the scan entry.

Figure 4.7: Patient Profile Page

The screenshot displays the 'Upload Scan for Patient' page in the PrivaFed application. The page header is identical to the previous screenshot. The main content area features a large teal background with a white form overlay. The form is titled 'Upload Scan for Patient' and contains the following fields: 'Scan File *' with a 'Choose File No file chosen' button, 'Scan Type' with a text input field containing 'e.g MRI, CT Scan', and 'Notes' with a text input field containing 'Optional notes'. At the bottom of the form, there are two buttons: 'Upload Scan' and 'Run Segmentation'.

Figure 4.8: Upload Scan Page

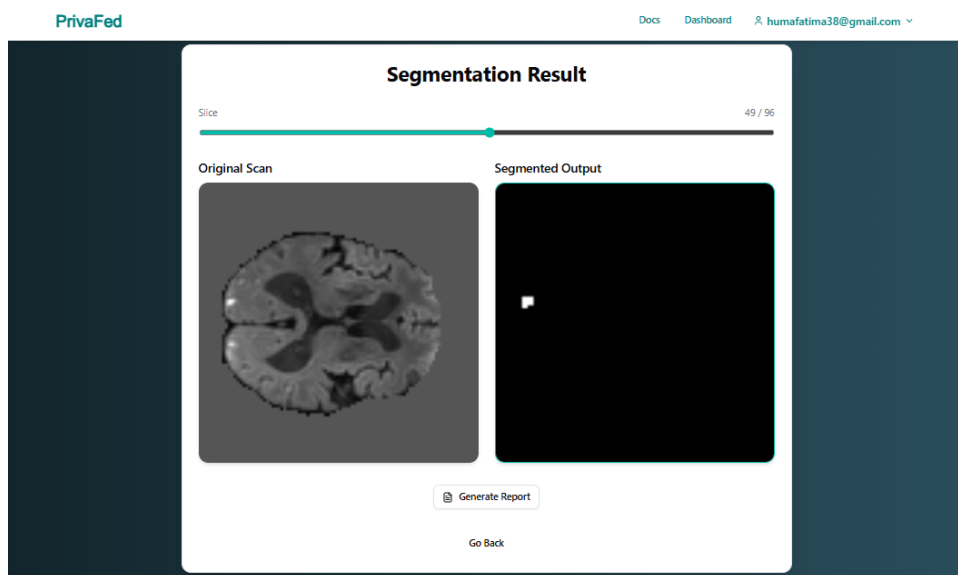


Figure 4.9: Segmentation Result Page

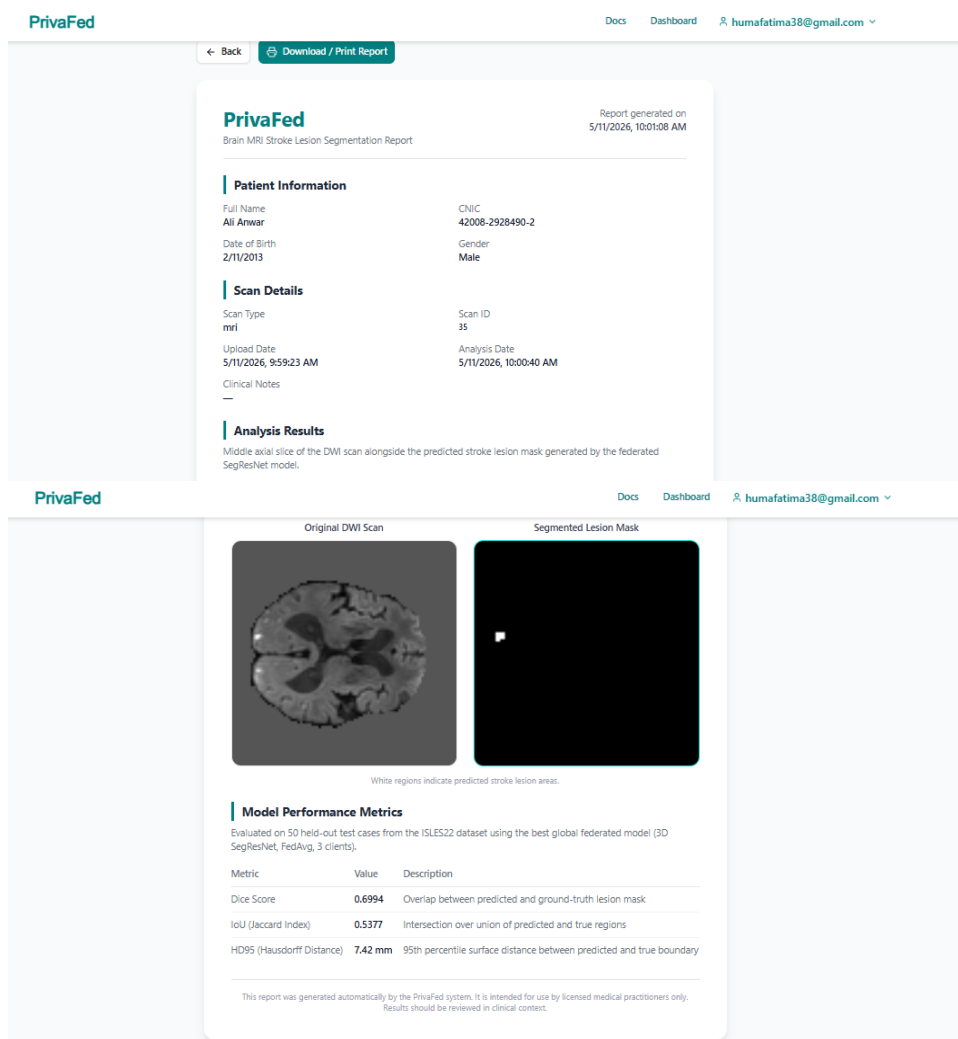


Figure 4.10: Download Report



Figure 4.11: Documentation Page

4.9 Database Design

This section includes the ER (Entity-Relationship) diagram and the data dictionary for the system. The ER diagram provides a visual representation of the data model, depicting the relationship between the entities. The data dictionary supports it by providing the types and descriptions of each data field in detail.

4.9.1 ER Diagram

The relationships among the entities Practitioner, Patient, Scan, and Report are illustrated in Figure 4.12. A practitioner can register a single account in the system and manage multiple patients under that account. For each patient, the practitioner can upload zero or multiple MRI scans. Each MRI scan generates exactly one report containing the segmented result. The segmentation results can be viewed directly within the application as well as be downloaded for further reference.

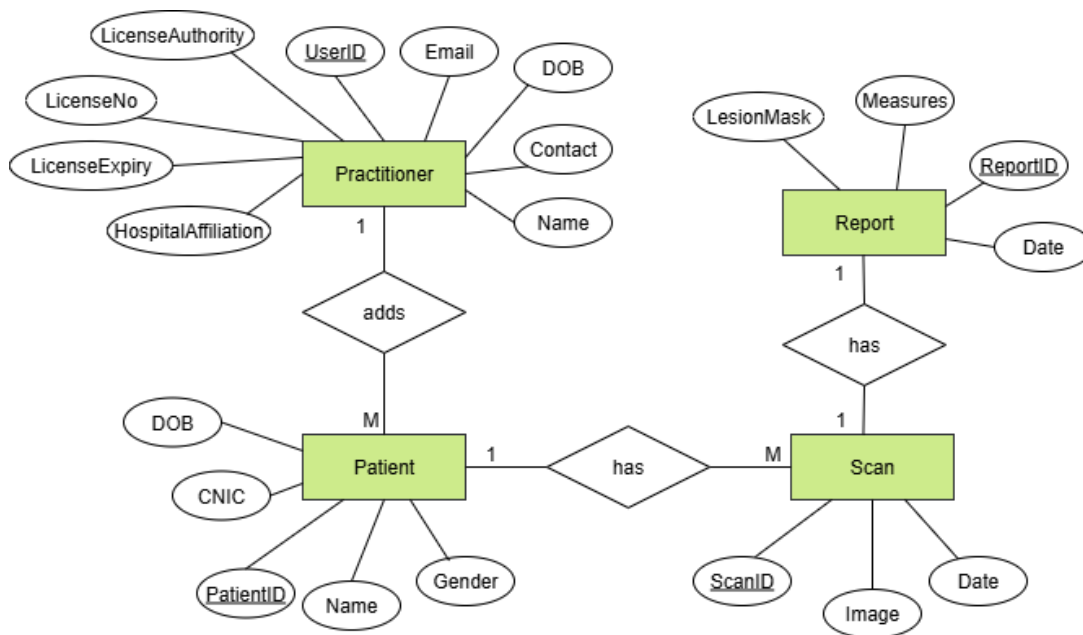


Figure 4.12: ER Diagram

Table 4.12: Data Dictionary

Entity	Attribute	Type	Nullable	Description
Practitioner	UserID (PK)	Integer	No	Unique identifier for each practitioner.
	Name	Varchar	No	Full name of the practitioner.
	Email (Unique)	Varchar	No	Practitioner's email used for login.
	Contact	Varchar	No	Phone number of the practitioner.
	HospitalAffiliation	Varchar	No	Hospital/clinic the practitioner is associated with.
	LicenseNo (Unique)	Varchar	No	Medical license/registration number.
	LicenseAuthority	Varchar	No	Medical authority who granted the license.
	LicenseExpiry	Date	No	Expiry date of practitioner's license.
Patient	PatientId (PK)	Integer	No	Unique identifier for each patient.
	UserID (FK)	Integer	No	Links patient to practitioner.
	Name	Varchar	No	Patient's full name.
	DOB	Date	No	Date of birth.
	Gender	Enum (M/F)	No	Gender of the patient.
	CNIC (Unique)	Varchar	No	National identity number of the patient.
Scan	ScanID (PK)	Integer	No	Unique identifier for each scan.
	PatientID (FK)	Integer	No	Links scan to the patient.
	Image	Varchar	No	Encrypted link to MRI scan image.
	Date	DateTime	No	Date and time the scan was uploaded.
Report	ReportID (PK)	Integer	No	Unique identifier for each report.
	ScanID (FK)	Integer	No	Links report to the scan.
	Date	DateTime	No	Date and time the report was generated.
	LesionMask	Varchar	No	Encrypted segmentation mask generated from scan.
	Measures	Varchar	No	Quantitative measures.

4.10 Risk Analysis

The project may confront one or more of the following risks:

4.10.1 Technical Risks

The development and deployment of the project bring along several technical risks. The quality of the MRI datasets is one of the potential problems. Noise, artifacts, and a lack of proper annotations can seriously affect the accuracy of segmentation. Another concern is the balance between performance and efficiency. Incorporating advanced approaches such as TL and FL may increase training time and require more computational resources if the model is not carefully optimized. Model generalization is also difficult to achieve because for it a framework must adapt well to heterogeneous datasets obtained

from different clients. This must be done without overfitting specific cases.

4.10.2 Operational Risks

The infrastructure required to support the system may pose operational hazards. Computational limitations may be imposed by low processing power or restricted GPU access. Model evaluation and training can be slowed considerably as a result. Moreover, synchronization between the clients and the central server may be hampered by erratic networks and other communication issues. Additionally, time constraints related to the FYP may limit the extent of experimentation and validation.

4.10.3 Business and Adoption Risks

There are adoption and economic risks associated with the system's acceptance and compliance in actual healthcare settings. Users may find it difficult to accept solutions based on AI for clinical decision making if they haven't been well validated. Regulatory compliance poses a risk as well because hospital deployment must adhere to the relevant authorities' norms and standards pertaining to medical data privacy.

4.10.4 Ethical Risks

A pivotal role is held by ethics in the field of medicine. A major risk is the possibility of a malicious user gaining access to the model and web interface. For instance, submission of incorrect or manipulated MRI scans could adversely affect the training process. As an outcome, the model will output inaccurate segmentation results. Such risks stress on the need for secure user authentication to ensure that only high quality data is used for model training.

4.11 Conclusion

This chapter outlined the complete SRS for the proposed system. This includes the system's features, functional and non-functional requirements, quality attributes, and associated risks. The mentioned specifications ensure that the system provides secure, reliable, and efficient stroke lesion segmentation while maintaining data privacy through FL. This section, therefore, serves as the basis for the design, development, and successful implementation of the PrivaFed framework.

Chapter 5 Proposed Approach and Methodology

Below are the details of the training process for PrivaFed's model development, aimed at stroke lesion segmentation. The implementation consists of data collection, preprocessing, and model training.

5.1 Data Input

The data to be used in the system will be obtained from open-source medical imaging datasets. Namely, ISLES'22, ISLES'24, and ATLAS will be used for model training and evaluation. The intended approach will use ISLES'22 and ATLAS for the initial training of the segmentation model. Next, the system will be shifted into a distributed environment to take advantage of FL. Finally, TL will be applied using ISLES'24, which is relatively smaller, to address data scarcity and improve the model's generalization across a wide range of edge cases.

5.2 Data Preprocessing

The MRI scans will first be reduced in size through lossless compression to maintain quality while allowing faster processing. Image normalization, filtering, noise reduction, and feature enhancement, will be applied to images to ensure variety and veracity within the dataset as additional steps. Together, these techniques ensure dataset consistency and prepare the input images for robust and reliable model training.

5.3 Training Process

The training process uses MRI data to develop initial models within a distributed environment. To improve performance, an FL framework is integrated for privacy-preserving training, while TL is brought in to solve the problem of limited data and strengthen the model's ability to generalize across different cases.

5.3.1 Segmentation Model

The major goal of the distributed model is to segment and mark stroke lesions accurately. Training will be conducted on freely available GPU resources such as Kaggle and Google Colaboratory. Experimentation with a number of deep learning architectures will be carried out. These include SegResNet, ResNet, Swin-UNet, nnU-Net, Swin Factorizer, and 3D UNet. The model demonstrating the highest accuracy and the best learning curve within a federated setup will be selected as the final architecture

for deployment.

5.3.2 Federated Framework

To tackle data privacy concerns, the system will adopt an FL framework. In this setup, training occurs locally across distributed nodes and only the model updates are aggregated centrally. This keeps sensitive patient data secure while allowing the model to improve through collaborative contributions from multiple clients.

5.3.3 Transfer Learning

After training the base model, TL will be used to enhance its adaptability. This approach allows the pretrained model to leverage its existing knowledge and adjust effectively to smaller datasets, such as ISLES'24. This helps in overcoming data scarcity, improving generalization, and enhancing accuracy in cases where the training data contains rare stroke lesion patterns or is limited in size.

5.3.4 Lightweight Approach

The clients will only send the encoder and bottle neck layers to the server for aggregation instead of the complete layers. The server aggregates the encoder and bottleneck layers to create global encoder and bottleneck representations. These layers will be distributed back to the client nodes allowing them to leverage from global encoder knowledge while still utilizing their own personalized decoders.

5.4 Conclusion

This chapter presented the proposed approach and methodology for stroke lesion segmentation. The central approach uses FL to guarantee data confidentiality and promote better model collaboration across multiple institutions. TL is then strategically used to enhance adaptability and performance on smaller and diverse datasets. By combining these techniques with the lightweight approach of partial layer aggregation, the proposed framework aims to achieve reliable, secure, and personalized stroke lesion segmentation.

Chapter 6 High-Level and Low-Level Design

In this chapter, the architectural and detailed designs of the proposed PrivaFed system are presented. The high-level design gives a general description of the major parts in the system and how they are interacting with each other whereas the low-level design gives the internal frameworks of modules, data movement, and implementation.

6.1 System Overview

PrivaFed is a privacy-preserving FL framework designed for brain stroke lesion segmentation using MRI scans. It enables various healthcare facilities to jointly train a common DL model without transmitting the patient data to guarantee data privacy. In addition to the federated training process, PrivaFed provides a user-friendly web application that enables authorized practitioners to upload MRI scans, view automated segmentation results, and generate structured analytical reports for clinical use. To summarize, the system integrates FL, TL, and web framework to deliver a secure and lightweight solution for medical image segmentation in real-world healthcare settings.

6.1.1 Overall Functionality

The system provides three core functionalities as listed below.

6.1.1.1 Federated Model Training

In PrivaFed, the federated model training process enables multiple clients, i.e., healthcare institutions, to collaboratively build a high-performing DL model for stroke lesion segmentation while simultaneously ensuring that no raw patient data leaves local premises. Every client site has its dataset and locally trains the shared model on its MRIs. Only the learned model parameters are sent to the central federated server instead of transferring patient data to the central location.

The central server performs secure aggregation, combining the received updates from all participating clients to form a refined global model. The aggregated model obtained is then redistributed back to the clients. This is followed by the next training round. This iterative process helps the global model progressively improve performance while ensuring privacy of local datasets. The federated training cycle continues until the model achieves a balance between data privacy, model accuracy, and communication efficiency.

6.1.1.2 Clinical Interaction via Web Interface

The web interface of PrivaFed is the source of interaction between the clinicians and the system. It provides an intuitive interface for the analyses of the uploaded MRIs. Further, it provides a secure mechanism for the authorized users (e.g. medical practitioners or health practitioners) to upload the MRI scans to the local client node without it being exposed to the central servers. Moreover, it allows visualization of result by overlaying the stroke lesions on the MRI scan allowing for easy interpretation. This allows the practitioners to create diagnostic reports which sums key findings such as affected brain regions and model confidence scores. Users is also able to add, delete and update attributes related to their patients, MRI scans and their results.

6.1.1.3 Personalization and Adaptation

After global model aggregation, PrivaFed adds additional steps to ensure model is relevant across various medical institutions and improves local performance. After each federated training round, fine tuning is performed on the globally aggregated model locally at each client node. This personalization and adaptation step added by PrivaFed allows the model to cater client-specific data characteristics, such as variations in MRI scanners and imaging protocols. This significantly improves model generalization and accuracy, especially on underrepresented and heterogeneous datasets.

6.1.2 Design Approach

The PrivaFed architecture is layered and modular, with each layer carrying out specific tasks that collectively guarantee secure operations, scalability, and maintainability. It is composed of three primary layers, namely Frontend, Backend, and FL.

6.1.2.1 Frontend Layer

The Frontend Layer consists of a web-based interface tailored for clinical use. Users can download structured reports, view the results of stroke lesion segmentation, and upload MRI scans. Since Interface focuses on simplicity, healthcare professionals can operate the system without any technical knowledge.

6.1.2.2 Backend Layer

The Backend Layer handles the requests between the frontend and the FL components and controls the flow of data from Supabase to the web interface. Through Supabase's Magic Link system, it combines access control and authentication to enable passwordless login for authorized users. This ensures that only verified clinicians can access sensitive information while providing a smooth user experience.

6.1.2.3 Federated Learning Layer

The FL layer makes up PrivaFed's core intelligence. It is in charge of data preprocessing, global aggregation, and secure model parameter exchange. To protect data privacy, only model updates are forwarded to the central server for aggregation.

6.1.3 Design Philosophy

The design philosophy of PrivaFed centers on three key principles, namely privacy, robustness, and usability.

6.1.3.1 Privacy

The system's privacy-first philosophy ensures that no unfiltered medical data is shared with parties other than the client institutions. Every model update is sent to the central server for compilation and this supports the upholding of patient privacy and adhering to clinical regulations on the protection of medical data.

6.1.3.2 Robustness

Robustness is achieved by an architecture when is capable of managing heterogeneous MRI datasets that includes data from different scanners and clinical scenarios. PrivaFed is made to function consistently even when there are changes in the format, size, and quality of the data.

6.1.3.3 Usability

Thirdly, the user-friendly interface simplifies interaction for clinicians by hiding the complexity of FL. Users can seamlessly upload scans, view segmentation results, and generate reports without the need for technical expertise because of PrivaFed's intuitive web interface.

6.2 Design Considerations

This section outlines the essential components that form the foundation of PrivaFed's design process.

6.2.1 Assumptions and Dependencies

This subsection discusses the fundamental assumptions and dependencies that are taken into account while designing the operational principles of the PrivaFed framework.

6.2.1.1 Assumptions

- All essential computational resources are available at every client site for local training and inference.
- There exists a stable and secure network connection between clients and the central aggregation server to allow smooth parameter exchange.
- Ethical and legal standards for medical data handling are strictly followed by each institution involved.
- It is the responsibility of the user to keep their login credentials safe to prevent unauthorized access to sensitive patient data.
- Despite being heterogeneous, client datasets contain sufficient diversity and volume, ensuring effective collaborative model training.
- Global aggregation server remains operational throughout the training process.

6.2.1.2 Dependencies

- The system relies on reliable and secure communication channels for the transfer of model parameters among clients and the central aggregation server.
- Local training, model updates, and aggregation tasks rely on DL frameworks like PyTorch and TensorFlow
- Backend services utilize Supabase for user authentication and secure session management via Magic Link login.
- GPU-enabled systems at client sites are necessary for efficient training of local models and reducing computation time.
- The web application requires a dependable hosting infrastructure to guarantee that the app is accessible consistently for the clinicians to upload MRI scans and receive segmentation results.

6.2.2 General Constraints

A number of global constraints shape PrivaFed's design, influencing its software architecture, performance, and deployment considerations.

6.2.2.1 Hardware Constraints

There is a probability of uneven computational resources at the different client sites involved, ranging from high-performance GPUs to standard CPU-based systems. This creates the need for adaptive training methods to balance workloads and guarantee equity among clients. This approach will help avoid performance bottlenecks during local training and model aggregation, resulting from hardware differences. It is crucial to use model optimization techniques to ensure efficient and synchronized federated training.

6.2.2.2 Network Constraints

Dependency on network stability is introduced for all participating clients when operating in a federated environment. Unstable connections and communication delays impact model synchronization and overall training efficiency. As a result, in order to serve clients with different network conditions, the system needs to support asynchronous updates. Furthermore, in order to minimize bandwidth consumption during parameter exchange, effective transmission protocols are required.

6.2.2.3 Data Privacy and Compliance Constraints

The system must fully comply with HIPAA (Health Insurance Portability and Accountability Act), GDPR (General Data Protection Regulation), and other such medical data protection regulations. Transmission of raw MRI data outside the client's local environment must not be allowed under any circumstances. Only encrypted model parameters or gradients are shared with the central server during training. This ensures that the system strictly follows the privacy-first approach.

6.2.2.4 Performance Requirements

Stable model convergence and consistent segmentation accuracy across all clients are requirements for the federated training process. To reduce latency during parameter exchange it is necessary to guarantee the effectiveness of communication between the clients and the central server. The system should guarantee optimized training rounds in order to balance performance with hardware and network limitations. During scan upload, result visualization and report generation, the web interface must give practitioners nearly real-time responses. This preserves a high level of responsiveness and computational efficiency.

6.2.3 Goals and Guidelines

Goals and guidelines that govern PrivaFed's design and development are discussed below.

- **Privacy and Security by Design:** To ensure data confidentiality, only the model parameters are

exchanged during the training process, and no patient data can be transferred out of the client institution.

- **Efficiency and Scalability:** The system will be made to be scaled easily and to offer resource efficiently. This will allow new institutions to join without major reconfiguration while supporting heterogeneous hardware environments.
- **User-Centric Design:** With user friendly and intuitive interface we ensure that clinicians can upload MRI scans, view segmentation results, and create structured reports. This helps conceal FL's complexity, allowing a seamless user experience.
- **Reliability and Fault Tolerance:** The system has asynchronous updates, retry procedures, and checkpoints to guarantee consistent operation in the event of a hardware or connectivity failure.
- **Ethical and Collaborative Development:** Led by transparency, fairness and accountability, the PrivaFed framework provides an ethical approach to AI and clinical partnerships to motivate trust, explainability and reliability in medical procedures.

6.2.4 Development Methods

PrivaFed employs an agile development strategy in its software design and development, prioritizing flexibility and incremental improvements. Although the method may not rigidly follow a specific formal framework, it is primarily influenced by the principles of Agile Scrum. This methodology offers both adaptability and quick responses to changing project requirements. Consequently, the system ensures compliance and resilience throughout its entire development life cycle.

6.3 System Architecture

The section diagrammatically provides the architectural description of the PrivaFed system. It provides a high-level overview of the system's functionality, as well as the decomposition of responsibilities among its components. The high-level architecture of PrivaFed is structured into four fundamental components: the web interface, the central server, the central database and the distributed client nodes.

Next.js and Tailwind is used to develop the web interface which gives healthcare professionals a secure access to the system. Using the dashboard, users can upload scans, look at the results of lesion segmentation and retrieve clinical reports. The federated aggregator is the central server which coordinates the global model updates by gathering local model parameters from clients. It then aggregates them and distributes the improved global model back to the clients. The central database is responsible for securely storing the user accounts, encrypted lesion masks while ensuring that raw MRI scans remain at

the client level. Finally, raw MRI scans and metadata are stored at the clients' (e.g., hospitals, clinics) local database. Clients perform training on their private data, and share only model changes with the central server to maintain privacy of their local dataset.

The combination of these elements guarantees privacy-preserving and effective workflow. The client-side training secures sensitive medical information, the server-side maintains global the model performance through effective aggregation of the local model updates and the web interface ensures ease and accessibility. The whole architecture is designed while considering robustness, scalability, and data security, which is in line with the healthcare requirements. The system architecture is illustrated in Figure 6.1.

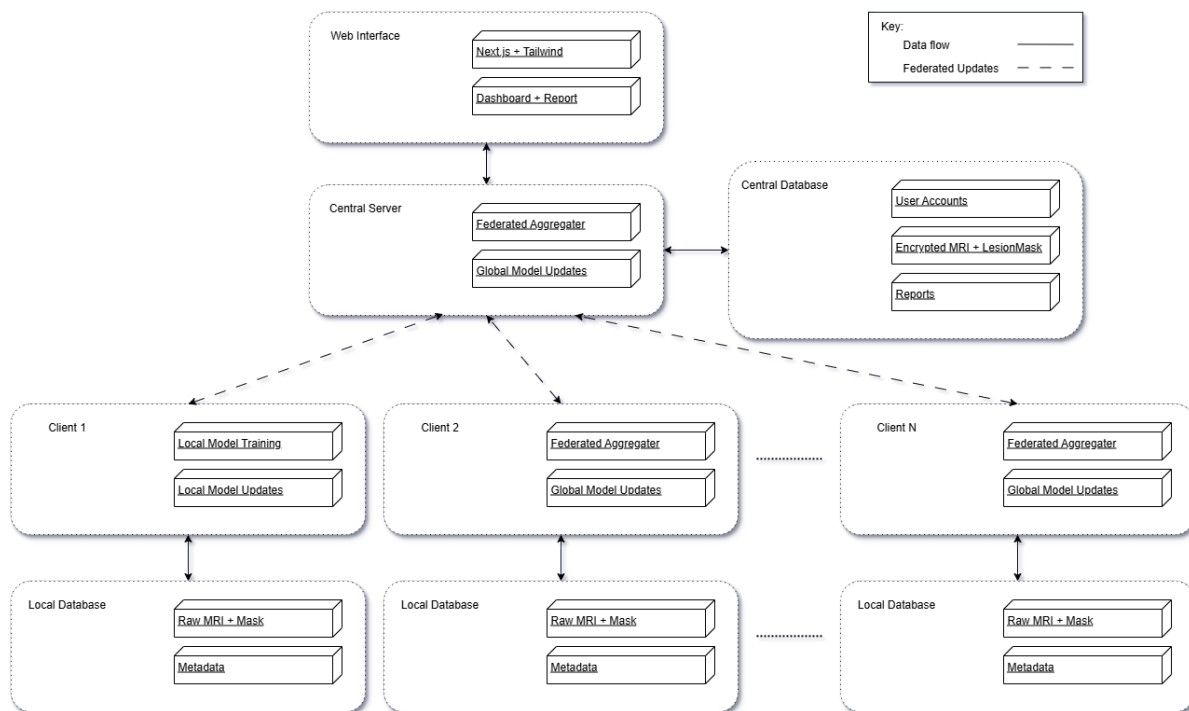


Figure 6.1: System Architecture Diagram

6.3.1 Central Server Subsystem

The central server subsystem is charged with the role of coordinating the FL process and guaranteeing safe collaboration among all involved clients. It collects model updates from hospitals, aggregates them and distributes the improved global model back to each client. The server side contain a secure database which stores encrypted lesion masks file path, reports and user access logs, but does not store any raw MRI scans, maintaining a certain level of data security.

Although our diagram represents a single central server for ease in understanding, in a real-world deployment, the federated aggregation service and application/database server may be separated to improve

modularity, scalability, and security.

6.3.2 Client Node Subsystem

The client node subsystem is located within each participating hospital and is involved in managing all sensitive medical data locally. Each client performs model training on its private dataset of MRI scans and produces lesion segmentation results without the transfer of raw data outside the hospital system. In order to maintain privacy, the central server only receives model weights or gradients during federated training. This architecture provides hospitals with complete control over their patient data while still benefiting from collaborative model improvement.

6.4 Architectural Strategies

PrivaFed employs various strategies to shape its architecture and the decision making processes.

6.4.1 Programming Languages

- **Python (Model Development):** Applied to the implementation of the FL and TL models, training, evaluating, and experimenting with the use of libraries, including PyTorch, TensorFlow, scikit-learn, NumPy, and Pandas.
- **Next.js (Web Application):** The full-stack React framework was selected to manage both the frontend user interface and the backend programming API routes. Next.js is made to simplify server-side rendering (SSR) and static site generation (SSG) and API endpoint generation, which is consistently efficient when integrating with the ML model and database.
- **TypeScript:** The statically typed superset of JavaScript is used to provide type safety, maintainability, and scalability of the web application.
- **Tailwind CSS and Shadcn UI:** Chosen for frontend styling due to their ability to facilitate rapid UI development with reusable and accessible components.

6.4.2 Database and Storage Strategies

Supabase has been chosen for real-time synchronization, metadata storage (scan IDs, report references), and secure user management via Magic Link authentication. Supabase is compatible with PostgreSQL, relational, and easily scalable for future use. Since only raw MRI scans are kept at the hospital level, the data storage policy protects patient privacy. Only reports, MRI scans, lesion masks, and a small amount of metadata are kept in the central system. Every image reference including MRI scan and lesion mask

is encrypted to prevent direct persistence of raw scans.

6.4.3 Web-Based Interface

PrivaFed uses a web-based interface that is user friendly and can be easily integrated into the working processes of the hospital. Magic Link Authentication is utilized for password-free login, which leads to less management problems and better security among the medical staff. The practitioners are able to upload MRI scans, see the results of segmentation, and download reports without the need of an additional software installation.

6.4.4 Distributed Learning and Networking

Every client node, which may consist of hospitals or clinics, trains a local model on its own local data. The client node shares only the model updates with the central aggregation server. This enables the server to refine the global model and distribute it back to the client nodes without accessing any raw MRI data. Such a FL setup ensures data privacy.

6.4.5 Extensibility and Future Enhancements

The system follows a modular structure with different components for pre-processing, local training, and reporting. Modular architecture allow easy updation and integration of new client nodes, algorithm or imaging modalities beyond MRI in the future.

6.5 Class Diagram

The relationships and interactions between the key elements of the PrivaFed system are defined by the following class diagram shown in Figure 6.2. The workflow revolves around the practitioners who manage patients, upload patient scans, and generate reports. Within the system, patients are stored as records rather than active users. Additionally, a report is generated for each scan to highlight the lesion segmented by the system.

6.6 Policies and Tactics

For PrivaFed, we focused on adopting tools that would make the system effective, easy to use, and efficient. These include selecting the right technologies, deciding on the best system architecture, and utilizing the best coding practices.

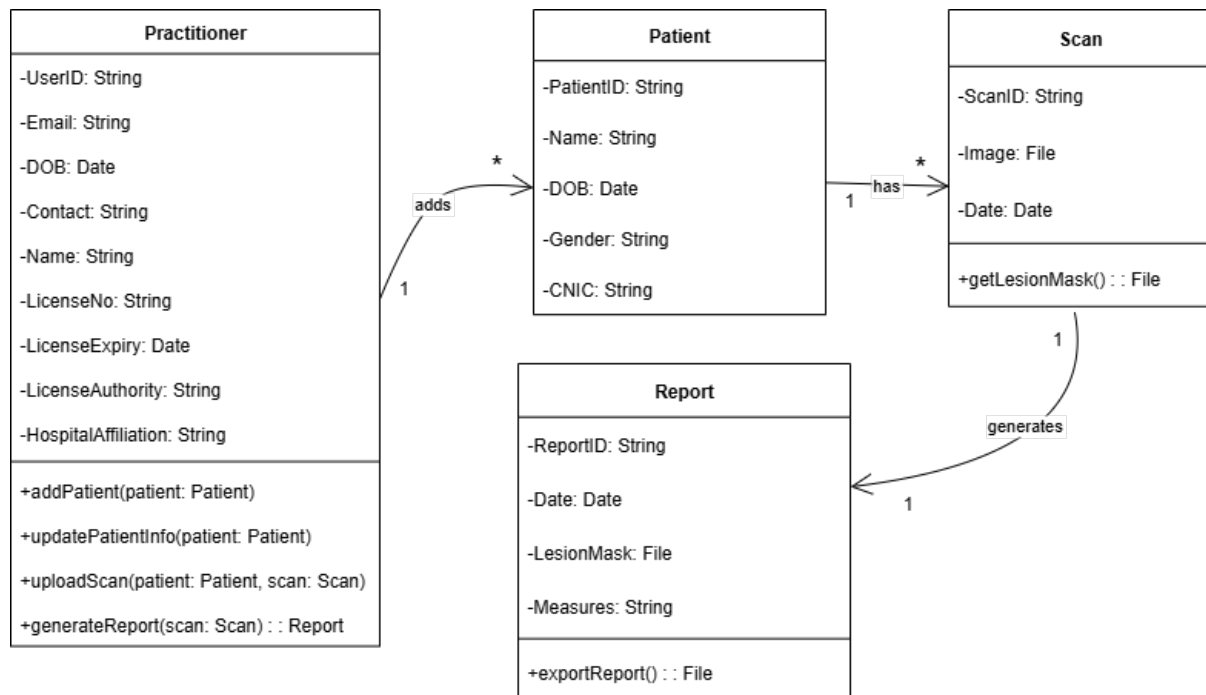


Figure 6.2: Class Diagram

6.6.1 Product Choices

The model development is based on Python due to its well-developed ML ecosystem with the involvement of libraries such as PyTorch, TensorFlow, NumPy, and Scikit-learn. For the web application, Next.js combined with TypeScript and Tailwind CSS, Shadcn UI offers a modern and responsive interface. Supabase was chosen to manage metadata, lesion masks, and reports, with MRI references stored in encrypted form for privacy.

6.6.2 Engineering Trade-offs

The project prioritizes privacy and robustness over raw system performance. Raw MRI scans for example are not ever centrally stored, which is a necessary condition to be in compliance with privacy-preserving goals, however this implies extra computation on the client side. Supabase was preferred for its built-in authentication and ease of integration, although it does not allow low-level customization.

6.6.3 Coding Guidelines and Conventions

Readability and maintainability is achieved by organizing code into separate directories such as frontend, backend, ml model. GitHub allows streamline development and deployment for easy version control. Additionally, README files and inline documentation enhance clarity and long-term sustainability.

6.6.4 Testing Policies

Various levels of testing will be utilized to ensure that application works correctly, performs well, and is free from critical bugs before reaching the users. The testing process will include unit tests for both front-end and back-end components, integration tests to cover the full workflow, and privacy validation tests to ensure that raw MRI data is never exposed at any processing stage.

6.6.5 End User Interfaces

The web interface is specifically tailored for healthcare professionals as the design is simple, and user-friendly. The design significantly focuses on readability, usability, and responsiveness that allow practitioners to upload MRI scans, view the outcome of lesion segmentation, and generate diagnostic reports with minimal effort.

6.7 Conclusion

The design of PrivaFed carefully strikes a balance between privacy protection, system resilience, and clinical usability. The system guarantees that private medical information stays in the client node while allowing for collaborative model improvement by fusing FL principles with a modular and scalable architecture. Technical efficiency and adherence to healthcare data standards are supported by the combination of contemporary web technologies and safe authentication techniques. PrivaFed provides a stable and long-lasting platform designed for practical clinical deployment and future scalability through systematic development procedures, comprehensive testing, and an intuitive user interface.

Chapter 7 Implementation and Test Cases

This chapter provides a detailed overview of the implementation process, algorithms used, and frameworks employed for the prototype of PrivaFed.

7.1 Implementation

The implementation mainly involved data acquisition, data preprocessing, and model selection.

7.1.1 Data Acquisition

PrivaFed leverages publicly available, open-source neuroimaging datasets on brain stroke. Datasets such as ISLES'22 and ISLES'24 were found to be particularly relevant and valuable for this study.

7.1.2 Data Preprocessing

These open-source datasets are already preprocessed to a considerable extent, leaving only a small amount of additional work required for PrivaFed. The main preprocessing step involved intensity normalization, and resampling to ensure that all 3D MRI scans are of the same size and optimized for effective training of the segmentation model.

7.1.3 Experimental Setups

PrivaFed's research phase included 3 different experimental setups.

7.1.3.1 ISLES'22

ISLES'22 contained 250 DWI samples and lesion masks. These were resized to (96,96,96) and were used in both centralized and federated setup.

- **Centralized Setup:** 80/20 training and validation split was utilized. 50 epochs were used while model training.
- **Federated Setup:** 3 client nodes were introduced, each having 50 samples for local model training. The global model was validated and tested on a validation and testing set of 50 samples each. 10 communication rounds and 20 local epochs were used for the federated setup.

7.1.3.2 ISLES'24

ISLES'24 contained 149 DWI samples and lesion masks. These were resized to (128,128,128) and were used in both centralized and federated setup.

- **Centralized Setup:** Training, validation and testing set consisted of 95, 24 and 30 samples and masks respectively. Model training took place over 50 epochs.
- **Federated Setup:** 2 client nodes were introduced, each having 50 samples for local model training. The global model was validated and tested on a validation and testing set of 20 and 29 samples respectively. The setup utilized 10 communication rounds and 20 local epochs.

7.1.3.3 Combined Experiment

Both ISLES'22 and ISLES'24 were used to create a non IID and heterogeneous setup in the federated environment. Both datasets were resized to (192,192,192). A total of 3 clients nodes were introduced as followed:

- **Client 0:** 100 samples of ISLES'24
- **Client 1:** 75 samples of ISLES'22
- **Client 2:** 75 samples of ISLES'22

For global validation 2 validation sets were utilized; ISLES'22 (50 samples) and ISLES'24 (24 samples). The global model was validated on both sets separately. Similarly, global testing also used 2 separate testing sets; ISLES'22 (50 samples) and ISLES'24 (25 samples).

The setup utilized 10 communication rounds and 20 local epochs. After obtaining the best global model, it was distributed back to the clients where each of them ran 5 extra local epochs to add personalization.

7.1.4 Model Selection

The primary focus of the development phase is to identify the best-performing model through trial and evaluation. Various models including Swin Factorizer, Swin UNETR, SegResNet and UNet are applied to the acquired datasets and assessed across multiple metrics such as Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and Hausdorff Distance (HD).

7.1.4.1 Centralized Setup

Initially, different models for stroke segmentation were explored, trained, and evaluated in a centralized environment to shortlist the most effective ones based on performance metrics.

7.1.4.2 Federated Setup

The models were transitioned into a distributed setup, which involved testing and evaluation to analyze performance under federated conditions. This included identifying the optimal aggregation technique,

determining which layers to include in aggregation, and assessing the impact of communication frequency and client participation ratio on overall model performance.

7.2 Test Case Design and Description

This section includes test cases to verify the functionality and performance of the system, along with the expected results and test outcomes.

Table 7.1: Test Case for Sign Up/ Sign In

Sign Up/ Sign In			
4.6.1			
Test Case ID:	1	QA Test Engineer:	Armeen Fatima
Test case Version:	1.0	Reviewed By:	Noor Ul Ain Asghar
Test Date:	12-04-2026	Use Case Reference(s):	4.6.1
Revision History:	None		
Objective:	To check if user is able to successfully sign up/ sign in to the website		
Product/Ver/Module:	Website sign up module		
Environment:	Website is running on web browser and internet is connected.		
Assumptions:	User has access to provided email and Magic Link service is properly integrated.		
Pre-Requisite:	User has a valid email address.		
Step No.	Execution description	Procedure result	
1	User selects "Sign Up / Sign In" option	Passed	
2	User enters a valid email address	Passed	
3	System generates and sends Magic Link	Passed	
4	User opens email and clicks Magic Link	Passed	
5	System validates the link	Passed	
Comments:	Test case passed. The system works according to our requirements.		
Passed			

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Table 7.2: Test Case for Logout

Logout			
4.6.2			
Test Case ID:	2	QA Test Engineer:	Armeen Fatima
Test case Version:	1.0	Reviewed By:	Huma Fatima
Test Date:	12-04-2026	Use Case Reference(s):	4.6.2
Revision History:	None		
Objective:	To verify that the user can securely log out and the session is properly terminated.		
Product/Ver/Module:	Website logout module		
Environment:	Website is running on web browser and internet is connected.		
Assumptions:	User is currently logged in and session management system is functional.		
Pre-Requisite:	User must be logged into the system.		
Step No.	Execution description	Procedure result	
1	User selects "Logout" option	Passed	
2	System terminates the session	Passed	
3	System redirects user	Passed	
Comments:	Test case passed. Logout functionality works correctly and securely.		
Passed			

Table 7.3: Test Case for Edit Profile

Edit Profile			
4.6.6			
Test Case ID:	3	QA Test Engineer:	Armeen Fatima
Test case Version:	1.0	Reviewed By:	Noor Ul Ain Asghar
Test Date:	13-04-2026	Use Case Reference(s):	4.6.6
Revision History:	None		
Objective:	To verify that the system successfully allows a user to edit and update their profile information		
Product/Ver/ Module:	User Profile Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and has access to the Edit Profile page		
Pre-Requisite:	User account exists prior		
Step No.	Execution description	Procedure result	
1	User navigates to Profile and clicks "Edit Profile"	Passed	
2	User updates valid fields and clicks "Save"	Passed	
Comments:	Test case passed. System successfully updates user profile and handles empty fields appropriately		
Passed			

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Table 7.4: Test Case for Add Patient

Add Patient			
4.6.7			
Test Case ID:	4	QA Test Engineer:	Huma Fatima
Test case Version:	1.0	Reviewed By:	Armeen Fatima
Test Date:	13-04-2026	Use Case Reference(s):	4.6.7
Revision History:	None		
Objective:	To verify that the system successfully allows an authorized user to add a new patient with valid information		
Product/Ver/Module:	Patient Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and has permission to add patients		
Pre-Requisite:	User has completely added the details and created a valid profile		
Step No.	Execution description	Procedure result	
1	User navigates to the Add Patient page	Passed	
2	User enters patient details and clicks "Save" button	Passed	
3	System saves the patient record successfully	Passed	
Comments:	Test case passed. System successfully adds new patients with proper validation		
Passed			

Table 7.5: Test Case for Search Patient

Search Patient			
4.6.8			
Test Case ID:	5	QA Test Engineer:	Noor Ul Ain Asghar
Test case Version:	1.0	Reviewed By:	Huma Fatima
Test Date:	13-04-2026	Use Case Reference(s):	4.6.8
Revision History:	None		
Objective:	To verify that the system successfully allows an authorized user to search patient record using valid/invalid inputs		
Product/Ver/ Module:	Patient Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and has permission to search patient records		
Pre-Requirement:	At least one patient record exists in the system		
Step No.	Execution description	Procedure result	
1	User enters a valid patient name in the search bar	Passed	
2	System displays list of records corresponding to searched name	Passed	
Comments:	Test case passed. System successfully searches for patients with valid input and handles invalid inputs		
Passed			

CHAPTER 7. IMPLEMENTATION AND TEST CASES

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Table 7.6: Test Case for Update Patient

Update Patient			
4.6.9			
Test Case ID:	6	QA Test Engineer:	Huma Fatima
Test case Version:	1.0	Reviewed By:	Noor Ul Ain Asghar
Test Date:	13-04-2026	Use Case Reference(s):	4.6.9
Revision History:	None		
Objective:	To verify that the system successfully allows authorized user to update existing patient records with valid information		
Product/Ver/ Module:	Patient Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and has permission to update patient records		
Pre-Requisite:	At least one patient record exists in the system		
Step No.	Execution description	Procedure result	
1	User selects a patient and clicks "Edit"	Passed	
2	User updates valid fields and clicks "Save"	Passed	
Comments:	Test case passed. System successfully updates patient records with valid data and handles invalid inputs		
Passed			

Table 7.7: Test Case for Delete Patient

Delete Patient			
4.6.10			
Test Case ID:	7	QA Test Engineer:	Noor Ul Ain Asghar
Test case Version:	1.0	Reviewed By:	Armeen Fatima
Test Date:	13-04-2026	Use Case Reference(s):	4.6.10
Revision History:	None		
Objective:	To verify that the system successfully allows an authorized user to delete a patient record		
Product/Ver/Module:	Patient Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and has permission to delete patient records		
Pre-Requisite:	At least one patient record exists in the system		
Step No.	Execution description	Procedure result	
1	User selects a patient and clicks "Delete"	Passed	
2	User confirms the deletion after deletion prompt shows	Passed	
Comments:	Test case passed. System successfully deletes patient records with proper confirmation dialog		
Passed			

Table 7.8: Test Case for Upload Brain Scan

Upload Brain Scan			
4.6.3			
Test Case ID:	8	QA Test Engineer:	Huma Fatima
Test case Version:	1.0	Reviewed By:	Noor Ul Ain Asghar
Test Date:	12-04-2026	Use Case Reference(s):	4.6.3
Revision History:	None		
Objective:	To verify that user can successfully upload a brain scan and the system processes it correctly.		
Product/Ver/Module:	Scan Management Module		
Environment:	Website is running on web browser with active internet connection.		
Assumptions:	User is logged in and has selected a patient. System supports valid image formats.		
Pre-Requirement:	User must be logged in and have a valid brain scan file.		
Step No.	Execution description	Procedure result	
1	User selects "Upload Scan" option	Passed	
2	User selects a valid brain scan file	Passed	
3	User uploads the brain scan	Passed	
4	System processes uploaded file	Passed	
Comments:	Test case passed. Upload functionality works correctly and handles errors appropriately.		
Passed			

Table 7.9: Test Case for View Analysis Results

View Analysis Results			
4.6.4			
Test Case ID:	9	QA Test Engineer:	Noor Ul Ain Asghar
Test case Version:	1.0	Reviewed By:	Armeen Fatima
Test Date:	12-04-2026	Use Case Reference(s):	4.6.4
Revision History:	None		
Objective:	To verify that the user can successfully view analysis results of an uploaded brain scan.		
Product/Ver/Module:	Scan Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in, patient is selected, and scan has been uploaded and processed		
Pre-Requisite:	A processed brain scan exists for the selected patient		
Step No.	Execution description	Procedure result	
1	User selects a patient from patient list	Passed	
2	User selects an uploaded brain scan	Passed	
3	User clicks on "View Results"	Passed	
Comments:	Test case passed. System correctly displays analysis results and handles in-progress scans appropriately		
Passed			

Table 7.10: Test Case for Generate Report

Generate Report			
4.6.5			
Test Case ID:	10	QA Test Engineer:	Noor Ul Ain Asghar
Test case Version:	1.0	Reviewed By:	Huma Fatima
Test Date:	12-04-2026	Use Case Reference(s):	4.6.5
Revision History:	None		
Objective:	To verify that the system successfully generates a detailed brain scan analysis report with segmentation results.		
Product/Ver/Module:	Report Generation Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in, scan is analyzed, and results are available		
Pre-Requisite:	A completed analysis result exists for the selected brain scan		
Step No.	Execution description	Procedure result	
1	User selects a processed brain scan	Passed	
2	User clicks "Generate Report"	Passed	
3	System generates report successfully	Passed	
4	User downloads or views report	Passed	
Comments:	Test case passed. System successfully generates and delivers brain scan reports with proper error handling.		
Passed			

Table 7.11: Test Case for View Patient History

View Patient History			
4.6.11			
Test Case ID:	11	QA Test Engineer:	Armeen Fatima
Test case Version:	1.0	Reviewed By:	Huma Fatima
Test Date:	13-04-2026	Use Case Reference(s):	4.6.11
Revision History:	None		
Objective:	To verify that the system displays all the scans for a specific patient		
Product/Ver/Module:	Scan Management Module		
Environment:	Web application running on browser with active internet connection		
Assumptions:	User is logged in and a patient has been successfully added		
Pre-Requisite:	A patient is selected		
Step No.	Execution description	Procedure result	
1	User navigates to the Patient Profile page	Passed	
2	System displays the patient record and scans successfully	Passed	
Comments:	Test case passed. System successfully displays all the scans of a patient.		
Passed			

7.3 Test Metrics

Table below shows the results of successful testing. All 11 test cases were executed and passed with no defects. The test set has a defect density of 0%, indicating that no issues are found. The test set

efficiency is 100%, reflecting the accuracy and completeness of the tests.

Table 7.12: Test cases Metric

Metric	Purpose
Number of Test Cases	11
Number of Test Cases Passed	11
Number of Test Cases Failed	0
Test Case Defect Density	0%
Test Case Effectiveness	100%
Traceability Matrix	-

7.4 Conclusion

This chapter described the implementation of PrivaFed and covered data collection, preprocessing, and model selection in centralized and federated environments. The development and execution of comprehensive test cases to evaluate system functionality and performance is also discussed in detail. The test results showed that all test cases completed successfully without any issues.

Chapter 8 User Manual

This chapter includes step-by-step instructions on accessing various features and performing key tasks of PrivaFed's application.

8.1 Features

- Sign Up/ Sign In
- Logout
- Edit Profile
- CRUD Patient
- Upload Brain Scan
- View Analysis Results
- Generate Report

8.1.1 Sign Up/ Sign In

The system uses a Magic Link instead of a password for user authentication. Upon entering the 'Sign Up/ Sign In' page, the user is asked to enter a valid email address. Upon submission the system sends a one-time Magic Link to the user's provided email address. The user then accesses their email and clicks on the Magic Link.

- If the link is valid:

A new account is created if the user is signing up. The user is logged in if the account already exists. The user is redirected to the dashboard.

- If an invalid email is entered:

The system displays an "Invalid email address" message.

- If the user does not receive the email:

The system provides an option to "Resend Magic Link".

- If the Magic Link is expired or already used:

The system displays "Link expired or invalid. Please request a new link."

8.1.2 Logout

To log out:

- Click the “Logout” button from the menu.
- Confirm your action if prompted.
- You will be redirected to the login screen, and your session will end.

8.1.3 Edit Profile

To edit your profile:

- Click the “Edit Profile” button.
- Update the desired fields and fill the empty fields
- Click “Save” to apply the changes.
- A confirmation message will appear once your profile has been updated successfully.

8.1.4 CRUD Patient

To add a patient:

- Click “Add Patient” button from the patient dashboard.
- You will be redirected to the add patient page where you will fill the form.
- Upon clicking “Save” button, you will be redirected to the patient dashboard and a confirmation message will appear.

To search a patient:

- Search for the desired patient name in the search bar.
- Results will be displayed below.

To update a patient:

- Click on the desired patient from the patient dashboard.
- You will be redirected to the patient profile page.
- Click “Edit” button and update the desired fields.
- Click “Save” to apply the changes and a confirmation message will appear.

To delete a patient:

- Click on the desired patient from the patient dashboard.
- You will be redirected to the patient profile page.
- Click "Delete" button and confirm your action if prompted.

8.1.5 Upload Brain Scan

- Click "Upload Scan" button from patient profile page.
- You will be redirected to the add scan page.
- Fill the required details in the form and click "Save" button.
- You will be redirected back to the patient profile page, and a confirmation message will appear.

8.1.6 View Analysis Results

- Navigate to the patient profile page.
- You click on the desired scan from the list.
- The system will display the analysis results, including segmentation findings for the scan.
- You can scroll through the results to review all detected regions and their details.

8.1.7 Generate Report

- After uploading the brain scan.
- Click the "Run Segmentation" button.
- The system will compile the analysis data and generate a detailed report.
- Once ready, click "Download" to save the report or "View" to preview it on screen.

Chapter 9 Experimental Results and Discussion

This chapter presents and analyzes the experimental results obtained from the implementation of the PrivaFed framework. The experiments aimed to assess the efficacy of privacy-preserving FL for medical image segmentation, particularly in detecting and segmenting stroke lesions in MRI scans. The models were evaluated using various subsets of publicly available datasets, including ISLES'22 and ISLES'24.

9.1 Centralized Setup

9.1.1 ISLES'22

All the models except for nnUNet were trained in the centralized setup for 50 epochs each using an 80/20 training and validation split. nnUNet utilized a 5-fold cross-validation technique, and each fold was trained for 20 epochs. Below is a brief overview of the results, accompanied by Table 9.1.

9.1.1.1 Swin Factorizer

Swin Factorizer is a hybrid model that combines Swin Transformer-based hierarchical feature extraction with factorized representations to efficiently capture both local and global contextual information. The model achieved a validation DSC of 0.6389 and a validation loss of approximately 0.2724.

9.1.1.2 Swin UNETR

Swin UNETR is a transformer-based segmentation model that integrates a Swin Transformer encoder with a U-Net-style decoder. It achieved a validation DSC of 0.6389 and a validation loss of approximately 0.2724.

9.1.1.3 nnUNet

nnUNet is a self-configuring deep learning framework that automatically adapts preprocessing, architecture, and training strategies. The model utilized 5 fold cross validation technique and achieved validation DSC of 0.7670, 0.7803, 0.7650, 0.8057, 0.6867 in each fold respectively. This gave an average DSC of 0.7609.

9.1.1.4 SegResNet

SegResNet is a convolutional neural network that incorporates residual connections into a segmentation framework to improve gradient flow and enable deeper, more robust feature learning. The Model

achieved a validation DSC of 0.6766 and a loss of 0.2732 appropriately. The final test DSC reached an approximate of 0.6068.

9.1.1.5 3D UNet

3D UNet is a convolutional neural network designed for volumetric data that uses 3D convolutions to capture spatial context across depth, height, and width. This U-shaped architectural model achieved a validation DSC and validation loss of 0.4182 and 0.3732 respectively.

9.1.1.6 2D UNet

2D UNet is a widely used encoder–decoder architecture. It achieved a best validation Dice coefficient of 0.6021 and a final validation loss of approximately 0.1949.

9.1.1.7 2D ResUnet

2D ResUNet combines the U-Net architecture with residual connections to enhance feature propagation and improve segmentation performance by mitigating vanishing gradient issues. The model successfully achieved a validation DSC of 0.6235 and a loss of 0.1821.

9.1.1.8 2D ResNet 34

ResNet-34 is a deep convolutional neural network with 34 layers that uses residual learning through skip connections to enable effective training of moderately deep architectures. After training, the model successfully achieved a validation DSC of 0.4814 and training loss of 0.5543.

9.1.1.9 2D ResNet 50

ResNet-50 is a deeper residual network with 50 layers that leverages bottleneck residual blocks to learn rich hierarchical features for improved representation. The model achieved 0.5824 and 0.4505 as validation DSC and train loss respectively.

Table 9.1: ISLES'22 Centralized Results

	Validation Metrics					
Models	Dice	IOU	HD95	Train Loss	Val Loss	Epochs
Swin Factorizer	0.6389	0.4694	14.4329	0.1477	0.2724	50
2D ResUnet	0.6235	0.453	12.1115	0.1203	0.1821	50
2D UNet	0.6021	0.4307	18.6133	0.1155	0.1949	50
3D UNet	0.4182	0.2644	23.8547	0.2713	0.3732	50
Swin UNETR	0.6379	0.4683	15.7341	0.1002	0.2097	50
2D ResNet 50	0.5824	0.4635		0.3915	0.4505	20
2D ResNet 34	0.4814			0.4000	0.5543	20
nnUNet (Fold 0)	0.7670					20
nnUNet (Fold 1)	0.7803					20
nnUNet (Fold 2)	0.7650					20
nnUNet (Fold 3)	0.8057					20
nnUNet (Fold 4)	0.6867					20
3D SegResNet	0.6766	0.5112	12.6028			50
	Test Dice: 0.6068	Test IOU: 0.4356	Test HD95: 14.2040			50

9.1.2 ISLES'24

After conducting centralized experiments on ISLES'22, the best two models; SegResNet and Swin UNETR, were brought into the centralized environment of ISLES'24. Training, validation and testing sets consisted of 95, 24 and 30 samples respectively. Model training took place over 50 epochs and the results have been summarized in Table 9.2.

Table 9.2: ISLES'24 Centralized Results

ISLES24	Validation Metrics					Test Metrics		
Model	Dice	IoU	HD95	Train Loss	Val loss	Dice	IoU	HD95
SegResNet	0.6247	0.4543	17.0105	0.4339	0.4333	0.6384	0.4689	15.1251
SwinUNETR	0.6087	0.4375	23.6302	0.5298	0.5314	0.628	0.4577	20.0223

9.2 Federated Setup

9.2.1 ISLES'22

In our federated learning setup, three clients trained local models on their own data while a central server aggregated the updates to form a global model. This enabled collaborative learning without sharing raw

data. Swin UNETR, Swin Factorizer, 3D U-Net, and 3D SegResNet were incorporated into the federated setting, as these models demonstrated strong performance in the centralized setup. Several aggregation techniques including FedAvg, FedProx, FedAdam, and Scaffold were tested. Initially, 5 local epochs with 50 communication rounds were used; however, this resulted in high communication costs and slow training. Using 20 local epochs with 5 rounds improved speed, but slightly reduced accuracy. The final setup with 20 local epochs and 10 rounds provided the best balance. Client classes handled local training and a server class managed aggregation and communication efficiently. The results have been summarized in Table 9.3.

Table 9.3: ISLES'22 Federated Results

		Test Metrics			Validation Metrics			
Models	Technique	Dice	IOU	HD95	Dice	IOU	HD95	Loss
Swin UnetR	FedAvg	0.6117	0.4407	13.7642	0.6605	0.4931	16.4493	0.28
	FedProx	0.6338	0.4640	13.5177	0.6674	0.5009	12.3521	0.2531
	FedAdam	0.4851	0.3202	31.4209	0.4528	0.2926	36.2403	0.7809
	Scaffold	0.6264	0.456	11.3586	0.6351	0.4653	16.7854	0.2704
Swin Factorizer	FedAvg	0.6580	0.4903	12.8081	0.701	0.5397	11.6846	0.2217
	FedProx	0.6494	0.4808	12.0627	0.6929	0.5301	12.5456	0.2269
	FedAdam	0.5216	0.3528	12.8669	0.5982	0.4267	12.4519	0.294
	Scaffold	0.6434	0.4742	13.9943	0.6763	0.511	13.8827	0.2395
3D Unet	FedAvg	0.4496	0.2900	15.0672	0.4884	0.3231	16.7481	0.4347
	FedProx	0.5122	0.3442	15.2051	0.5396	0.3695	18.1925	0.4167
	FedAdam	0.0412	0.0210	60.3952	0.0232	0.0117	63.9858	0.8972
	Scaffold	0.4609	0.2995	17.5051	0.5166	0.3482	16.2832	0.3542
3D SegResNet	FedAvg	0.6994	0.5377	7.4197	0.7488	0.5984	9.4671	0.1962
	FedProx	0.7196	0.5620	6.6449	0.7503	0.6004	10.3695	0.1952
	FedAdam	0.5058	0.3385	17.8105	0.4629	0.3012	20.4921	0.7866
	Scaffold	0.6853	0.5212	8.6609	0.7406	0.588	11.0562	0.1954

9.2.2 ISLES'24

After conducting extensive experiments on ISLES'22 in federated environment, only the best aggregation techniques; FedAvg and FedProx, were tested on ISLES'24 in the distributed environment. Among

the four models tested as shown in Table 9.4, SegResNet and Swin UNETR gave the best performance across both aggregation techniques.

Table 9.4: ISLES'24 Federated Results

ISLES24		Validation Metrics				Test Metrics		
Technique	Model	Dice	IoU	HD95	Val loss	Dice	IoU	HD95
FedAvg	SegResNet	0.7232	0.5664	11.325	0.2099	0.7106	0.5511	15.671
	SwinUNETR	0.6912	0.5281	14.7168	0.3736	0.6439	0.4748	25.6699
	Swin Factorizer	0.695	0.5326	17.3251	0.2291	0.6236	0.453	28.9228
	3D UNET	0.5124	0.3445	40.2078	0.8129	0.5326	0.3629	46.5695
FedProx	SegResNet	0.6818	0.5172	11.2785	0.2671	0.6916	0.5285	18.6065
	SwinUNETR	0.6919	0.529	18.5717	0.3174	0.6504	0.4819	29.6636
	Swin Factorizer	0.6758	0.5104	22.794	0.2452	0.603	0.4317	27.7165
	3D UNET	0.6346	0.4648	23.6152	0.4267	0.6421	0.4729	27.3342

9.3 Combined Experiment

The best model, SegResNet, was used in the combined experiment. It was experimented with the top two aggregation techniques; FedProx and FedAvg. Hence the following 2 experiments were conducted:

1. **SegResNet with FedProx** shown in Table 9.5 and 9.6.
2. **SegResNet with FedAvg** shown in Table 9.7 and 9.8.

In total 3 clients nodes were used. 2 of the clients involved contained ISLES'22 dataset while 1 client contained ISLES'24 dataset. This setup was conducted to test imbalance, heterogeneous and non IID datasets.

After creating the global model via the central server, it was distributed back to the clients where each trained the global model extra 5 local epochs on their local dataset to add in personalization. Adding personalization allowed the global model to be adapted to the local dataset while still achieving a DSC of above 0.66, which is still better than what was achieved in centralized setup.

Table 9.5: SegResNet (FedProx) - Global Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 22 - Global	0.7204	0.5629	20.4579	0.2101	0.6749	0.5094	19.3545
	ISLES 24 - Global	0.7487	0.5983	12.0232	0.1822	0.6785	0.5135	15.4096

Table 9.6: SegResNet (FedProx) - Personalized Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 22 - Client 1	0.7194	0.5618	21.6714	-	0.6801	0.5153	16.3888
	ISLES 22 - Client 2	0.6718	0.5058	25.6434	-	0.6709	0.5047	21.7772
	ISLES 24 - Client 0	0.7361	0.5824	13.4383	-	0.645	0.476	13.3411

Table 9.7: SegResNet (FedAvg) - Global Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedAvg)	ISLES 22 - Global	0.7319	0.5772	17.2905	0.2017	0.6842	0.52	16.4383
	ISLES 24 - Global	0.7542	0.6053	13.2494	0.1767	0.6317	0.4616	16.4045

Table 9.8: SegResNet (FedAvg) - Personalized Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedAvg)	ISLES 22 - Client 1	0.7214	0.5642	18.7428	-	0.6795	0.5145	13.8927
	ISLES 22 - Client 2	0.6996	0.5379	22.3642	-	0.6568	0.4889	18.2882
	ISLES 24 - Client 0	0.7302	0.575	10.501	-	0.6023	0.4309	21.0357

9.4 Lightweight

To make the FL framework more lightweight and communication-efficient, only the encoder layers, including the bottleneck layer, of the SegResNet model were transmitted to the server for aggregation instead of sharing the complete model parameters. This reduced the communication overhead and computational cost associated with transferring large model weights between the server and client nodes by 6.41%. The full model consisted of 4,700,897 parameters, while the lightweight model contained 4,399,344 parameters.

The server aggregated the encoder and bottleneck layers from all participating clients, via FedProx to generate global feature representations. These aggregated layers were then redistributed back to the client nodes, allowing them to leverage globally learned encoder knowledge while still maintaining their own personalized decoder layers for local specialization. Retaining personalized decoders enabled each client to better adapt to the characteristics of its local dataset while still benefiting from collaborative learning across heterogeneous and non-IID datasets.

The lightweight experiment was conducted using 3 different setups:

1. **ISLES'22** shown in Table 9.9.
2. **ISLES'24** shown in Table 9.10.
3. **Combined Experiment** shown in Table 9.11.

Results of the experiments are summarized below:

Table 9.9: ISLES'22 Setup Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 22 - Client 0	0.7368	0.5833	13.339	0.2013	0.6848	0.5207	10.8743
	ISLES 22 - Client 1	0.7119	0.5527	9.5745	0.2184	0.662	0.4947	9.8931
	ISLES 22 - Client 2	0.7293	0.5739	13.3703	0.2074	0.6754	0.5099	8.5568

Table 9.10: ISLES'24 Setup Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 24 - Client 0	0.7092	0.5494	13.098	0.2165	0.6600	0.4925	22.2648
	ISLES 24 - Client 1	0.6983	0.5365	17.0628	0.2313	0.6824	0.5179	24.937

Table 9.11: Combined Setup Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 22 - Client 1	0.6836	0.5193	16.579	-	0.663	0.4959	16.2062
	ISLES 22 - Client 2	0.6823	0.5178	19.8099	-	0.6661	0.4993	17.7804
	ISLES 24 - Client 0	0.7199	0.5623	12.8392	-	0.6516	0.4833	17.6038

9.5 Transfer Learning

In the TL experiment, the best-performing model, SegResNet, and best-performing aggregation technique, FedProx, was utilized. Initially, the model was pretrained on the ISLES'22 dataset to learn generalized stroke lesion segmentation features. The pretrained weights obtained from ISLES'22 were then distributed across the client nodes through the central server. The clients further trained the model on the ISLES'24 dataset for local adaptation. This approach enabled the model to transfer the knowledge learned from ISLES'22 while adapting to the variations present in ISLES'24, including differences in data distribution and imaging characteristics.

Experiments:

1. **TL from ISLES'22 to ISLES'24:** The final model achieved a test DSC of 0.7064, demonstrating reasonable lesion segmentation performance on the ISLES'24 dataset as shown in Table 9.12. It was able to achieve best validation DSC in the third communication round, showing that it took advantage for the pretrained weights and converged quickly.
2. **TL from ISLES'22 to ISLES'24 and lightweight:** The federated ISLES'24 setup was used to evaluate the effect of transferring knowledge from ISLES'22 to ISLES'24 while aggregating only the encoder and bottleneck layers. With the lightweight aggregation strategy, the highest DSC values for both clients were achieved at the fourth communication round, reaching 0.7109 and 0.7228, respectively as shown in Table 9.13. Although a slight decrease in performance was observed on the test set, with DSC values of 0.6984 and 0.7051 for Clients 0 and 1, respectively (average DSC of 0.7017), the combination of TL and lightweight aggregation still produced strong segmentation performance while reducing the amount of information exchanged during federated training.

Table 9.12: Transfer Learning Results

Model	Validation Metrics				Test Metrics		
	Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	0.7315	0.5766	12.6526	0.2013	0.7064	0.5461	16.3068

Table 9.13: Transfer Learning and Lightweight Results

Model	Dataset	Validation Metrics				Test Metrics		
		Dice	IoU	HD95	Loss	Dice	IoU	HD95
SegResNet (FedProx)	ISLES 24 - Client 0	0.7109	0.5514	10.1506	0.2685	0.6984	0.5365	14.3489
	ISLES 24 - Client 1	0.7228	0.5659	10.9345	0.2486	0.7051	0.5445	17.6264

9.6 Conclusion

To conclude, this chapter has shared and discussed the results captured by implementing the PrivaFed framework. Starting from a centralized environment and shifting to a distributed federated setup, this chapter gives an overview of numerous experiments conducted to find the best overall technique and model to be utilized in the final product.

Chapter 10 Conclusions

PrivaFed addresses the growing need for privacy friendly, collaborative, and effective AI solutions on brain stroke lesion segmentation. In this study, we explored the integration of FL and TL to develop a decentralized yet high performing framework suitable for real-world clinical applications.

A deeper understanding of the problem domain was developed, including stroke characteristics, lesion variability, and challenges in medical image segmentation. A comprehensive literature review was conducted, analyzing state of the art architectures such as UNet, SegResNet, Swin Transformer based models, and nnU-Net, as well as techniques like such as TL and hybrid approaches to improve generalization from limited datasets.

Several models were evaluated in the ISLES'22 centralized setting. Among 2D architectures, ResUNet obtained the highest validation DSC of 0.6235. Among the 3D architectures, nnU-Net achieved the highest DSC 0.7609, and SegResNet also showed a strong and competitive performance with DSC of 0.6766, demonstrating its effectiveness and stability in volumetric medical image segmentation tasks. Transformer based models such as Swin UNETR and Swin Factorizer also showed competitive performance, highlighting the importance of attention mechanisms in medical imaging tasks.

Based on these results, a FL setup was implemented on ISLES'22 dataset. Several 3D architectures, including Swin Factorizer, Swin UNETR, SegResNet, and U-Net, were trained across distributed clients. Different configurations of local epochs and communication rounds (e.g., 5 local epochs with 50 rounds, and 20 local epochs with 5 or 10 rounds) were tested to study the trade off between communication efficiency and model convergence. Setting 20 local epochs with 10 rounds provided the best balance.

Furthermore, several aggregation techniques including FedAvg, FedProx, FedAdam, and SCAFFOLD were explored on the ISLES'22 experimental setup. In particular, the best federation performance was achieved when combining SegResNet and FedProx with a Dice score of 0.7196. This highlighted both the robustness of SegResNet and the effectiveness of FedProx in handling data heterogeneity across clients. These experiments showed that FL can achieve performance comparable to centralized approaches while maintaining data privacy.

The next phase of our research included experimenting on ISLES'24 dataset. Only two models, SegResNet and Swin UNETR, were evaluated in the centralized setting, giving a DSC of 0.6384 and 0.628 respectively. However, in the federated setting for ISLES'24, four models and two aggregation techniques were tested to draw results.

The third phase of our research included experimentation on heterogeneous, imbalanced, and non IID dataset. A combined experiment was carried out using SegResNet with two different aggregation tech-

niques; FedProx and FedAvg. This experiment also tested the element of personalization, by allowing each client to adapt the global model to their local dataset by running the final global model for 5 extra epochs on their local dataset.

The fourth phase implemented a TL approach by utilizing pretrained weights from ISLES'22 for the ISLES'24 segmentation task. This enabled the model to leverage previously learned stroke lesion representations, improving convergence and segmentation performance, with SegResNet and FedProx achieving a test Dice Score of 0.7064. The model was able to achieve the best validation DSC in only 3 communication rounds with the help of TL. Initially, it took 5 communication rounds.

Finally, a lightweight personalized FL approach was also explored, where only the encoder and bottleneck layers were shared globally while each client retained its own decoder. This reduced communication overhead by 6.41%, while maintaining competitive performance, with Client 0 and Client 1 achieving Dice Scores of 0.6984 and 0.7051.

To improve ease of use, a web application was developed, that allows secure upload of MRI scans and real-time visualization of segmentation results. This provides a practical interface for medical practitioners while maintaining a focus on privacy and accessibility.

The objectives of this study were mostly achieved, including evaluating segmentation models, implementing a FL framework, and developing a functional system for privacy-preserving analysis. The results show that FL can maintain strong performance while ensuring data privacy. Additionally, incorporating more diverse datasets and MRI modalities can further improve model generalization and clinical applicability.

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